

# Rural Migrants and Urban Informality: Evidence from Brazil

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## Abstract

This paper studies the economic effects of rural-urban migration on Brazilian cities. Using a shift-share IV design, we show that, over a decade, drought-induced immigration reduces informality, has no effect on unemployment, and increases the number of formal firms and jobs. Downward formal wage adjustments play a key role, as these long-run effects are weaker in regions with stronger wage rigidity. In the short run, when wage rigidity is strongest, we replicate the informality-increasing effects documented in the literature. We develop and estimate a model of firm dynamics and informality that rationalizes these results. The counterfactuals reveal that, in the short run, the informal sector absorbs the expanding labor force and acts as a “stepping-stone” to formality for firms and workers. In the long run, however, it reduces the aggregate benefits from immigration by allowing the least productive firms to survive.

*Keywords: Informality, rural-urban migration, labor markets, firm dynamics*

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# 1 Introduction

The urban population in developing countries grew by 12.5 percent between 2015 and 2020, and it is projected to grow by 64.7 percent by 2050 (UNCTAD, 2021). Rural-urban migration flows account for a substantial fraction of this population growth (Jedwab et al., 2017), which are likely to intensify due to climate change (Rigaud et al., 2018). Whether urban developing economies will be able to generate enough good jobs to accommodate this fast-growing workforce is a fundamental question for economic development (The World Bank, 2013).

The available empirical evidence on the *short-run* effects of rural-urban migration provides reasons for pessimism (El Badaoui et al., 2017; Kleemans and Magruder, 2018; Corbi et al., 2021). These studies largely confirm the traditional “Harris-Todaro-Fields” view (Harris and Todaro, 1970; Fields, 1975), which posits that immigration increases unemployment or informality because downward nominal wage rigidity prevents formal sector job creation. This perspective aligns with a broader literature on worker- and firm-side frictions in developing countries’ urban labor markets, which predominantly focuses on short-term employment effects of economic or policy shocks (e.g. Quinn and Woodruff, 2019; Caria and Orkin, 2024).

We investigate the *long-run* economic effects of rural-urban migration on Brazilian cities. Contrary to existing literature, we show that immigration *reduces* informality, has no impact on unemployment, and increases the number of formal firms and jobs over a decade. Downward nominal wage rigidity (DNWR) in the formal sector emerges as a key mechanism. Over the long run, immigration exhibits weaker formalization effects and generates higher non-employment in regions with stronger baseline DNWR, consistent with the “Harris-Todaro-Fields” framework. In the short run – when DNWR is most binding – we replicate the informality-increasing effects documented in previous studies using year-on-year specifications. We develop and estimate a model of firm dynamics and informality that rationalizes these empirical patterns. The counterfactuals reveal that the informal sector initially absorbs additional labor supply, while simultaneously serving as a stepping stone to formality during the transition to a new steady state. However, in the long run, the informal sector reduces the aggregate economic benefits of immigration.

In the first part of the paper, we combine matched employer-employee data on the universe of formal firms and workers in Brazil between 1995 and 2018, with

individual level data from three waves of the Demographic Census (1991, 2000 and 2010). We use these data and a shift-share IV design to identify the causal effects of immigration on local labor market outcomes and formal firm dynamics at destination. We focus on decadal changes in outcomes between 2000 and 2010. To construct our shift-share instrument, we use detailed information on previous migration patterns across municipalities to construct the “shares”. To construct the “shifts”, we combine granular data on land use, growing seasons, and a monthly measure of dryness in each municipality based on the Standardized Precipitation-Evapotranspiration Index (SPEI, see [Vicente-Serrano et al., 2010](#)).

We find that internal immigration in Brazil between 2000 and 2010 had no effect on unemployment, and *increased* the share of formal employment: an increase in the immigration rate of one percentage point increases the share of workers in formal wage employment by 0.27 percentage points (a 1.2 percent increase). This effect is entirely driven by a shift of workers from informal to formal jobs, with no change in wage employment overall, unemployment, or self-employment. Wages in the formal sector fall by 0.7 percent while informal wages are unaffected. We find similar results if we look at migrants and non-migrants or high- and low-skilled workers separately. The latter is consistent with the fact that immigration induced by droughts does not change the skill composition of the workforce at destination in our setting.

We turn to the effects of internal immigration on formal firms’ outcomes. We find that an increase in the immigration rate by one percentage point leads to a 1.6 percent increase in the number of firms and a 2 percent increase in the number of formal jobs. There is also greater churn, with more formal firms entering and exiting every year (2.5 and 2.7 percent, respectively). We show that these new firms are not created by the migrants themselves. When we examine effects over time, we find that they remain constant or increase slightly throughout the 2010’s (until 2018).

We carry out a number of robustness checks to alleviate concerns about identification. Reassuringly, we do not find evidence of differential pre-trends, and we assess various potential threats to the exclusion restriction. In particular, we conduct several checks to ensure that our results are not driven by other channels than migration, such as excluding agricultural firms or firms that process agricultural products, to controlling for drought shocks locally and in nearby municipalities, and for exposure to capital reallocation from rural areas via the bank network ([Albert et al., 2021](#)). As a final and key robustness check, we construct an alternative shift-share instru-

ment in which the shifts are a combination of land use with international commodity price shocks.<sup>1</sup> We find very similar results, strengthening the credibility of our main empirical results, which are not dependent of the push shocks we use.<sup>2</sup>

The result that rural-urban migration decreases informality in urban labor markets stands in stark contrast with the predictions from [Harris and Todaro \(1970\)](#) and [Fields \(1975\)](#) and the available empirical evidence ([Kleemans and Magruder, 2018](#); [El Badaoui et al., 2017](#); [Corbi et al., 2021](#)). Downward nominal wage rigidity (DNWR) in the formal sector is central to the “Harris-Todaro-Fields” framework, because it prevents the formal sector to expand following an immigration shock. DNWR is also important in our context: nominal wage cuts are actually illegal in Brazil, and the minimum wage increased regularly in the 2000’s ([Engbom and Moser, 2022](#)). We investigate the role of DNWR in two ways. First, we calculate the municipality-level Kaitz index as the ratio between the national minimum wage and municipalities’ median wages in 2000. Second, we measure DNWR as the share of workers with negative year-on-year hourly nominal wage changes among workers with non-zero wage changes ([Rodríguez-Clare et al., 2024](#)). Our results show that in municipalities with high DNWR the formalization effects of immigration are smaller or even completely muted, and non-employment increases. These findings therefore echo the “Harris-Todaro-Fields” view that when formal wages are rigid rural-urban migration leads to higher urban unemployment and informality.

DNWR is more likely to hold in the short than in the long run, since inflation and worker turnover allow formal firms to adjust real wages downward. This could explain why we find that rural-urban migration fosters formalization over a decade, while previous work shows that it increases informality or unemployment in the following year ([El Badaoui et al., 2017](#); [Kleemans and Magruder, 2018](#)). In a final empirical exercise, we directly assess this conjecture by estimating the year-on-year specification of [Corbi et al. \(2021\)](#). We find the same patterns documented in previous work: in the short run, rural to urban migration leads to an increase in informality driven by a decline in formal wage employment.

In the second part of our analysis, we develop a new model of firm dynamics and informality that is able to rationalize these surprising findings. It extends the

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<sup>1</sup>The construction of the shock is similar to [Imbert et al. \(2022\)](#) for China.

<sup>2</sup>The drought and the price shocks are completely independent across origins, and the two shift-shares are only weakly correlated at destination (0.3). Compliers are also different: price-induced migrants are more likely to be young and male compared to drought-induced migrants and natives.

canonical model of firm dynamics of [Hopenhayn \(1992\)](#) in two ways. First, we include the two margins of informality considered in [Ulyssea \(2018\)](#): (i) whether to register their business or not, the extensive margin; and (ii) whether firms that are formally registered hire their workers with or without a formal contract, the intensive margin. Modeling the intensive margin is key, because it links formal firms and migrants even if they only take up informal jobs when arriving in cities.

Second, we allow for heterogeneous growth profiles (similar in spirit to [Sterk et al., 2021](#)), which implies that informality can have opposing effects in the economy. On the one hand, informality can act as a “stepping-stone” to formality for both firms and workers, as it introduces greater *de facto* flexibility in the presence of burdensome regulations and rigidities in the formal sector. However, it also allows less productive firms to survive and compete with more productive formal firms, weakening the natural selection process in the economy. The net effect of these forces and their interaction with immigration shocks is an empirical question.

We calibrate the model and use it to perform counterfactual simulations of the effects of a once-and-for-all 10 percent labor supply increase, which corresponds to the 80th percentile of the migration shocks we see in the data. The counterfactual results confirm the main findings from the long-run, instrumental variable analysis: the supply shock leads to a small reduction in the share of informal workers, a reduction in wages and a sizable increase in the number of formal firms. Importantly, the *magnitudes* of the effects are close to those we estimate using the IV method for an immigration shock of the same size.

Furthermore, the counterfactual results show that 40 percent of the increase in the number of formal firms comes from higher formalization of informal firms throughout their life cycle. This highlights the importance of accounting for firm dynamics, as a static model would not capture this mechanism and could therefore substantially underestimate the formalization effects. Output per worker declines, which is largely explained by compositional effects, as the share of formal firms and workers at the bottom quartiles of the firm productivity distribution increases.

We then examine the transition dynamics between steady states assuming that there is downward wage rigidity in the formal sector. We show that immediately after the labor supply shock, both labor and firm informality increase. As formal wages start to slowly adjust downwards, firm informality responds almost immediately and starts declining. The share of informal labor hovers around its initial level for a few

years before converging to its lower level in the new steady state. This is due to the intensive margin of informality, as the new formal firms tend to be small and hire a substantial fraction of their labor force informally.

Our final counterfactual combines the labor supply shock with higher enforcement intensity on the extensive margin of informality, nearly shutting down the informal sector. We do not see this as an actual policy counterfactual but rather as a thought experiment that sheds light on the role of the informal sector in absorbing this shock. We show that under stronger enforcement and lower informality, migration has stronger positive effects on output and firm productivity in the long run, at the cost of a substantial displacement of the least productive informal firms.

This result highlights the different short- and long-run consequences of informality. In the short run, the informal sector helps absorb the increased labor force following a migration shock and acts as a “stepping-stone” to formality for some of these firms and workers. This role is more important when formal wage rigidity is stronger. In the long run, however, the informal sector dampens the economic benefits from immigration by allowing the least productive firms in the economy to survive.

Our paper is relevant to four strands of the literature. Our main contribution is to highlight the stark difference between short- and long-run impacts of rural-urban migration on urban labor markets in developing countries. The extensive literature has focused on the former, growing around the idea of wage rigidity in the formal sector ([Harris and Todaro, 1970](#); [Fields, 1975](#)). Consistent with this view, the empirical literature documents negative, short-run employment effects of immigration for resident workers, especially in the formal sector ([Kleemans and Magruder, 2018](#); [El Badaoui et al., 2017](#); [Corbi et al., 2021](#)).<sup>3</sup> In contrast, we document a reduction in informality, and an increase in the number of formal firms and jobs over a decade. Our reduced-form evidence by low- and high-DNWR labor markets, combined with the model-based counterfactuals, consistently show that sluggish wage adjustment in the formal sector can explain these differences between short- and long-run effects.

Second, our findings provide new insights to the broader discussion on job creation in developing countries. Most of the literature takes labor demand in the formal sector as given, and documents frictions that make it hard for job-seekers to enter the formal sector ([Alfonsi et al., 2020](#); [Abebe et al., 2021](#); [Carranza et al., 2022](#)). Our

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<sup>3</sup>There is also a direct link between undocumented international migration and informality, which is not relevant in our context ([Bahar et al., 2021](#); [Elias et al., 2018](#)).

empirical results show that a labor supply shock can generate demand side responses and lead to formal employment creation. We build on the literature that studies firms dynamics and informality (e.g. [Dix-Carneiro et al., 2024](#); [Erosa et al., 2022](#)) to develop a new model that can rationalize this channel. Our contribution is to introduce a richer entry structure and heterogeneous growth profiles, which allows the informal sector to act as a “stepping-stone” into formality. This is key, as 40% of the formalization effects come from informal firms that eventually formalize.

Third, our paper relates to a handful of papers that causally estimate the economic effects of internal migration in developing countries beyond labor markets ([Imbert et al., 2022](#); [Albert et al., 2021](#)) Our contribution is to combine empirical and model-based approaches to investigate the relationship between internal migration and the urban economy, and to shed light on the role of wage rigidity in shaping short- and long-run effects. We show that immigration generates formal sector expansion but decreases productivity per worker in the long run.

Finally, we contribute to the literature on population growth and firm dynamics. Our findings are a mirror image of the recent literature on demographic decline and the “start-up deficit” in the US, which argues that the demographic slowdown had negative effects on firm entry, labor reallocation, employment and firm growth ([Hopenhayn et al., 2018](#); [Karahan et al., 2019](#); [Pugsley and Sahin, 2019](#); [Peters and Walsh, 2022](#)). Our findings also relate to the growing literature on immigration and firms in developed economies ([Lewis, 2011](#); [Peri, 2012](#); [Olney, 2013](#); [Kerr et al., 2015](#); [Mitaritonna et al., 2017](#); [Dustmann and Glitz, 2015](#)) We bring some of the first evidence on the effects on formal firms dynamics, formal-informal worker allocation, and the role of formal wage rigidity in a developing country context.

The paper is structured as follows. Section 2 presents the data and Section 3 the shift-share instrumental variable empirical design and its results. Section 4 develops the model, while Section 5 discusses the calibration and counterfactual results.

## 2 Data

This section describes the data sources we use to estimate the causal effects of immigration on the local economy (Section 3), and to calibrate our model (Section 5).

**Migration and Labor Market Outcomes** The first dataset we use is the Decennial Population Census, which contains information on individuals’ socioeconomic characteristics and labour market outcomes. Crucially, the Census also contains detailed information about individuals’ migration patterns. In our analysis, we focus on the last two waves of the Census – 2000 and 2010 – and restrict the sample to working-age adults (15 to 64 years old).

Our unit of analysis is the Minimum Comparable Area (MCA), which combines municipalities whose borders have changed during the study period: there are 3,545 such MCAs in our final sample. We will call them “municipalities” for simplicity. Following the census data structure, we define as migrant a person who came to their current municipality of residence in the last ten years, and compute the cumulative immigration flows between 2000 and 2010 for each municipality pair. We focus on flows to urban locations, as defined in the census (88% of all migration between 2000 and 2010), and across state borders (40% of migration to urban areas between 2000 and 2010). We then compute the immigration rate in each urban destination as the sum of immigration flows over the decade divided by the population in 2000. This is our main endogenous regressor in the instrumental variable analysis (Section 3). Internal immigration in the average urban destination is large: 17.6% overall, 7% for state-to-state migration (Panel D in Table 1). Figure 1 shows the geographical variation in immigration rates across municipalities, which is one of the main sources of variation used in our empirical analysis.

Additionally, we use the census to compute socio-demographic characteristics and labor market outcomes for each destination municipality, which are shown in Table 1. The main socio-demographics (Panel A) are the share of female, of young (below 18) and high-skilled (completed secondary education and above) in the working age population. Our main labor market outcomes are the share of working-age adults employed in the private sector as their main occupation, which we split between formal and informal wage employment. In Brazil, formal employees are required to have a “signed work booklet” (*carteira de trabalho assinada*), and the Census directly asks wage workers whether they have it. We categorize formal workers as those who have this booklet, and informal otherwise.<sup>4</sup> We compute the average real monthly wage for all private sector workers, and separately for formal and informal workers (we compute hourly wages as well for some robustness checks). As Panel B in Table 1

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<sup>4</sup>See for example, [Ponczek and Ulyssea \(2022\)](#) for a description of labor regulations in Brazil.



shows, formal private wage work expanded in Brazil between 2000 and 2010, while informal wage work stagnated. Wages increased by 11% (6% in the formal sector).

Finally, to capture year-on-year variation in urban labor market outcomes we use the National Household Survey (PNAD), which is conducted annually by the National Bureau of Statistics (IBGE) also responsible for the Demographic Census. It is only possible to identify 700 municipalities in the PNAD, which nevertheless account for 65 percent of the employed population in Brazil.<sup>5</sup> Even though the PNAD is not representative at the municipality level, it provides comparable municipality-level labor market outcomes to those obtained in the Census (see [Corbi et al., 2021](#)). We use data from 2001 to 2009 for these 700 municipalities to compute the same labor market outcomes obtained from the Census.

**Firms** The third data set we use is the *Relação Anual de Informações Sociais* (RAIS), which is a matched employer-employee administrative dataset from the Ministry of Labour in Brazil that contains the universe of formal firms and workers. We use these data for the period of 1997-2018. For the instrumental variable analysis, we use the RAIS data to compute moments related to firm dynamics at the municipality level (Panel C in Table 1). In particular, we compute the logarithm of the total number of formal establishments and formal jobs, and of the number of establishments that entered and exited in the last year. Entry in the data can be due to firm creation or formalization of an establishment that was previously operating informally. Exit is characterized when a given establishment is no longer found in the administrative data. We also calculate the real average formal wage in each municipality by dividing the monthly wage by the number of hours worked and an inflation index and averaging it over all formal employees. As Panel C in Table 1 shows, the 2000s have seen tremendous growth in the number of formal firms (40%) and workers (60%).

We use additional data on informal firms and informal workers within formal firms to compute the relevant moments we need for the model calibration. We thus complement the RAIS data with the ECINF survey (*Pesquisa de Economia Informal Urbana*), a cross-section representative of all Brazilian firms with up to five employees, which was also collected by IBGE in 2003.<sup>6</sup> This is a matched employer-employee

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<sup>5</sup>We are very grateful to [Corbi et al. \(2021\)](#) for providing us with the municipality codes that allow us to identify these municipalities in the PNAD data.

<sup>6</sup>The data include firms with up to 10 employees, but the information for firms with more than five employees is not representative ([Ulyssea, 2018](#)).

data set that contains information on entrepreneurs, their businesses and employees. In particular, it includes information about the businesses' age (months/years in operation), their formalization status and of their employees. Thus, the data allow us to capture both the extensive and the intensive margins of informality.

**Push Shocks** We use a shift-share IV design to identify the causal effect of migration on destination outcomes (see Section 3 ahead). To construct our instrument, we need exogenous shocks that push people to emigrate from their home municipalities. We consider the occurrence of droughts as the main push shock.

To measure dryness in each municipality in each month we use the Standardized Precipitation-Evapotranspiration Index (SPEI), which measures the water balance based on precipitation and evapo-transpiration due to temperature.<sup>7</sup> The data are available on a geo-referenced grid which we match to Brazilian municipalities. We construct an indicator for drought which corresponds to negative values of the SPEI for each month  $m$  in each municipality of origin  $o$  ( $D_{om}$ ). We then combine the micro data of the 1980 Agricultural Census with information on the growing season of each crop in a given region to build an indicator  $g_{ocm}$ , which equals one if the crop  $c$  is growing in municipality  $o$  in month  $m$ . Finally, we create the drought shock as the combination of the drought indicator in the growing season of each crop, with weights equal to the share of each crop in the value of agricultural production in that municipality,  $\pi_{oc}$ . In our main specification using decadal changes, we accumulate all shocks throughout the decade as follows:

$$s_o^{drought} = \sum_m \sum_c (\pi_{oc} \times g_{ocm} \times D_{om}) \quad (1)$$

As a robustness check, we use variation in agricultural incomes due to changes in international crop prices as an alternative push shock following Imbert et al. (2022).<sup>8</sup> Interestingly, the price and the drought shocks are completely independent across origins (correlation of 0.007). Figure 2 shows the geographic distribution of the drought shock, while the map of the price shock can be found in the Appendix B.1.

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<sup>7</sup>The SPEI has been built by climate scientists Vicente-Serrano et al. (2010) and can be freely downloaded here <https://spei.csic.es/home.html>. It has been used in economics by Bertoli et al. (2020) and Albert et al. (2021), among others.

<sup>8</sup>Appendix B.1 describes the construction of the price shock in detail.

### 3 Instrumental Variable Analysis

#### 3.1 Empirical Design

We estimate the impact of immigration on labor markets outcomes by regressing the change in outcome  $y$  in municipality  $d$  between 2000 and 2010, on the immigration rate over the decade  $Mig_d$ :

$$\Delta y_d = \beta_0 + \beta_1 Mig_d + \beta'_2 \mathbf{X}_d + \varepsilon_d \quad (2)$$

where  $\Delta y_d = y_{d,2010} - y_{d,2000}$  is the long difference in labor market outcomes: employment rate in the private wage sector (formal and informal) and log wages in the private sector (formal and informal). The vector  $\mathbf{X}_d$  includes socio-demographic controls: the share of female, young and high-skilled workers in municipality  $d$  in 2000. We report robust standard errors, and use 2000 population as regression weight.

We then turn to firm outcomes, for which we have annual data between 1997 and 2018. We use a specification similar to Equation 2 and regress changes in the number of formal firms, entrants and exiting firms, in the number of formal jobs and in the firm wage on the migration rate, controlling for socio-demographic controls. To reduce noise and increase stability of the entry and exit measures (especially in very small municipalities), we compute outcomes as two-year averages. For comparability with the labor market results, we first focus on the decadal changes between 1999-00 and 2011-12. We then exploit the full extent of the panel data and estimate the effects of migration on changes in outcomes between 1999-00 and 2011-12, 2013-14, 2015-16 and 2017-18. We also investigate the presence of pre-trends, and use as a dependent variable the change in outcomes between 1997-98 and 1999-00.

**Identification** Since we estimate all regressions in first differences, our specification implicitly accounts for municipality fixed effects. By including controls, we also allow for differential trends across municipalities with different initial socio-demographic conditions. However, we may not identify the causal effect of immigration on labor markets and firms due to reverse causation and omitted variable bias: for example, cities with thriving labor markets are likely to attract more migrants. To overcome these threats to identification, we rely on a shift-share instrumental variable design, in which the shifts are assumed to be exogenous (Adão et al., 2019; Borusyak

et al., 2022). More specifically, we combine cross-sectional variation across destination municipalities in their pre-existing migration networks with different origins (the “share”), and time variation in exogenous push shocks that affect migration incentives at origin (the “shift”). Formally, the instrument writes:

$$Z_d = \sum_o \lambda_{o,d} s_o^{drought} \quad (3)$$

where  $Z_d$  denotes the instrument for immigration into the municipality of destination  $d$ , based on the drought shock  $s_o^{drought}$  to the municipality of origin  $o$  described in section 2 (Expression 1).  $\lambda_{o,d}$  denotes the share of migrants from origin  $o$  among migrants who had come at destination  $d$  between 1995 and 2000.<sup>9</sup> We then use  $Z_d$  as an instrument for  $Mig_d$  in a 2SLS estimator.

Our design depends on two key identification assumptions. First, baseline migration shares may be endogenous to future outcomes, but the push-shocks (shifts) are as good as randomly assigned (Adão et al., 2019; Borusyak et al., 2022). Second, the exclusion restriction holds, and the instrument only affects the dependent variables via the immigration shares. We present ample evidence in support of these assumptions in our robustness checks analysis in Section 3.4.

Importantly, we also use an alternative shift-share instrument in which the shifts are generated by international price shocks and check that we obtain similar results. Given that the shifts are independent across origins and the two shift-share instruments are only partly correlated across destinations (correlation of -0.3), this tests whether our results are driven by specific pairs of sending and receiving municipalities.

## 3.2 Effects on Labor Markets

We start by discussing the effects of internal immigration on urban labor markets presented in Table 2. The OLS estimates suggest that immigration correlates positively with formal wage employment and negatively with informal employment, with a small positive correlation with overall wage employment and wages (insignificant). One would expect the correlation with (formal) wage employment and wages to be biased upwards since migrants would be attracted by destinations with better labor

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<sup>9</sup>The 2000 Census only contains retrospective information on migration going 5 years back, hence we compute the migration shares from 1995 to 2000.

markets outcomes. We next turn to the IV estimates, which have a causal interpretation.<sup>10</sup> The coefficient on wage employment turns negative but insignificant (see Column 1), while the effect on formal employment becomes even more positive and the effect on informal employment more negative (Columns 2 and 3). Quantitatively the estimates suggest that a one percentage point increase in the migration rate – which corresponds to 18.5 percent of a standard deviation – increases formal employment by 0.27 percentage points (a 1.2 percent increase from the mean of 23 percent) and decreases informal employment by 0.28 percentage point (a 2.8 percent decrease from the mean of 10 percent). Since the overall effect on wage employment is zero, the increase in formal sector work is driven by a shift away from informal employment.<sup>11</sup> Turning to Columns 4 to 6 (Table 2), we find evidence that immigration slightly decreases formal wages. A one percentage point increase in the immigration rate reduces wages in the formal sector by 0.7 percent, with no effect on informal wages. The overall effect on wages is null, which could be the result of a positive compositional effect – as workers shift from the low paying informal sector to the formal sector – that counteracts the decline in formal wages. It could also be the result of an increase in labor demand in the long run, due to capital / labor substitution or a shift towards labor intensive products and production processes, which we cannot observe in our data but have been highlighted by Imbert et al. (2022) in China.<sup>12</sup>

The literature has emphasized the importance of looking at the labor market response to immigration separately by skill, especially when migrants have different skill levels than natives (Dustmann and Glitz, 2015). Panel B of Appendix Table A.4 shows that immigration flows induced by droughts do not change the labor force composition at destination. Consistently, Appendix Table A.5 shows very similar effects on employment and wages for high- and low-skilled workers. Finally, we check in Appendix Table A.6 that our results are the same among migrants and non-migrants, although wage effects are more negative for non-migrants.

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<sup>10</sup>Table A.1 in the Appendix presents first stage estimation results.

<sup>11</sup>In Appendix Table A.2 we show that there are no effects on non-employment, self-employment, or non-remunerated work, but there is a small decline in the fraction of workers who report being employers and domestic workers.

<sup>12</sup>Although we use the main occupation in the last month and monthly wages as our main outcomes, we check in Appendix Table A.3 that the results are similar when we use the share of hours spent in each occupation and hourly wages instead. Informality shows a stronger reduction that is not completely compensated by an increase in formal hours, which is reflected in a very small decline in wage employment. This could be a result of workers reducing their workload to comply with the labor regulation as they formalize and shift away from the unregulated informal sector.

### 3.3 Effects on Formal Firms

We now turn to firm-level data to study the effect of immigration on decadal changes in the number of formal firms, entry, exit, number of jobs and firm average wages. Panel A in Table 3 present the OLS estimates, while Panel B presents the IV results. Comparing the OLS and IV results, the effects on number of firms, entry, exit and number of jobs remain positive and if anything increase in magnitude. Our estimates indicate that a one percentage point increase in the immigration rate leads to a 1.6 percent increase in the number of firms and a 2 percent increase in the number of formal jobs. There is also higher churn, with a larger number of firms entering and exiting every year, 2.5 and 2.7 percent for a one percentage point increase in the immigration rate, respectively. The negative point estimate for average wages confirms the worker-level findings of a decrease in formal wages (also with similar magnitude), but the estimate is not significant.

To put these effects in perspective, the start-up deficit documented in the U.S. corresponds to a decline of 5 percentage points in firm entry between 1980 and 2012 (Pugsley and Sahin, 2019). Hence, the labor supply shock induced by rural to urban migration has a first order effect on firm dynamics. This effect does not seem to be driven by migrants creating firms at higher rates: Table A.7 in the Appendix shows that migrants are not more likely to be business owners in municipalities that receive more immigration. This is true for small firms with less than 5 employees, and larger ones with at least 6 employees.

We can also use the time dimension of the RAIS data to study the effects of migration on formal firms beyond 2011-12. Specifically, we estimate the effects of migration on changes in outcomes for different time windows: between 1999-00 and 2011-12 (our main results), 2013-14, 2015-16, and 2017-18. As Figure 3 shows, the effects tend to be somewhat stable or slightly increasing over time. Reassuringly, the figures do not show any evidence of pre-trends, with small and insignificant IV coefficients in the period before immigration took place (1997-98).

### 3.4 Robustness

We start by checking if pre-trends may be a concern, which Figure 3 already suggests it is not the case. To formally check that pre-trends do not drive our results, we run the benchmark specification controlling for lagged changes in outcomes and show that

our results remain essentially unchanged (Panel A of Appendix Tables [B.1](#) and [B.2](#)).

We perform a number of additional robustness checks, which we report in Panels B–E in Appendix Tables [B.1](#) and [B.2](#). Another potential threat to our instrumental variable strategy would arise if migration rates are very persistent overtime, so that we are capturing the effects of previous migration waves, a concern which has been raised in particular by [Jaeger et al. \(2018\)](#). To alleviate this concern, we control for log population at baseline in Panel B, and for migration rates between 1995 and 2000, in Panel C. Finally, we allow for differential trends by baseline GDP and industry composition (both computed in 2000) in Panels D and E, respectively. Our results are robust to these controls. In Appendix Tables [B.3](#) and [B.4](#) we implement a transformation from destinations to origins and run the regressions at the origin level as suggested by [Borusyak et al. \(2022\)](#). In addition to the benchmark OLS and IV regressions transformed at the origin level (Panels A and B, respectively), we also estimate the reduced-form specification (Panel C). The inference procedure in this specification is asymptotically equivalent to that of [Adão et al. \(2019\)](#), and therefore it alleviates concerns about inference in our main specification.<sup>13</sup>

Although we interpret our IV estimates as the effects of immigration, droughts could affect firms via other channels, which would violate the exclusion restriction. First, shocks to rural incomes caused by droughts may affect the demand for goods produced by urban firms ([Santangelo, 2016](#)). In Panel A in Appendix Tables [B.5](#) and [B.6](#), we show that our results do not change after we control for drought shocks in the municipality of destination and the sum of shocks in other municipalities weighted by the inverse of distance. Second, droughts could induce capital reallocation towards migration destinations. To capture this channel, we construct an alternative shift-share that combines drought shocks at origin with bank links (deposits and loans) between all municipalities, as in [Albert et al. \(2021\)](#). Our results remain unchanged when we add it as a control (Panel B in Appendix Tables [B.5](#) and [B.6](#)). Third, drought shocks may have indirect effects on urban firms that process agricultural goods. To alleviate this concern, we exclude from the sample agricultural firms, and manufacturing firms that process food and beverages or tobacco, and our results are unchanged (Panel C in Appendix Table [B.6](#)).

As a final robustness check, we construct a second instrument based on shocks to

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<sup>13</sup>[Adão et al. \(2019\)](#) show that conventional standard errors can be invalid in frameworks such as ours, as observations with similar exposure shares can have correlated residuals.



rural incomes coming from variations in international crop prices (see Section 2 and Appendix Section B.1 for more details). Appendix Tables B.7 and B.8 show similar results, albeit with stronger effects on firm entry and exit, formal and informal wages. These results are important, as they show that our main results are not driven by specific pairs of sending and receiving regions: the price and drought shocks are independent across origins (correlation of 0.007), and the two shift-shares are only weakly correlated across destinations (around 0.3). They also suggest that our results are not due to a specific set of migrant workers.<sup>14</sup>

### 3.5 Mechanisms

To a large extent, the rural-urban migration literature has grown around the idea that internal migrants join the ranks of casually employed or unemployed workers in urban areas. This idea dates back at least to Harris and Todaro (1970) and Fields (1975), and has been supported by recent empirical evidence (El Badaoui et al., 2017; Kleemans and Magruder, 2018; Corbi et al., 2021). Our results run counter to this dominant view, as they show that internal migration increases formal employment, reduces informality, and reduces formal wages, with no effect on unemployment.<sup>15</sup>

A prominent assumption in the theoretical frameworks of Harris and Todaro (1970) and Fields (1975) is Downward Nominal Wage Rigidity (DNWR) in the formal sector. DNWR is essential to explain that the formal sector does not accommodate migrant workers, who instead become unemployed or who are hired by informal firms that are not constrained by DNWR. In the context of our study, Brazil in the 2000s, perfect DNWR was imposed by law in the formal sector: nominal wage cuts were not allowed. There was also a binding national minimum wage that was increased regularly (Engbom and Moser, 2022). By definition, informal employment relationships were not subject to these constraints.<sup>16</sup>

Importantly, DNWR is more likely to hold in the short than in the long run. With inflation, real wages in the formal sector can adjust downward, so that formal

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<sup>14</sup>Panel A of Appendix Table A.4 shows that unlike droughts, price shocks induce migration flows that change the composition of the workforce at destination: they bring in younger male workers.

<sup>15</sup>Imbert et al. (2022) also find negative wage effects and positive employment effects of internal immigration focusing on large manufacturing firms in China over five years.

<sup>16</sup>There has been increasing empirical evidence on the importance of DNWR coming from micro data (e.g. Grigsby et al., 2021; Hazell and Taska, 2023), as well as renewed interest in the role of DNWR in the macro, trade and labor literatures (Rodríguez-Clare et al., 2024).



employment can increase. This could explain why we find that rural-urban migration increases urban formality over a decade, while previous empirical papers find that it increases urban informality or unemployment in the following year (El Badaoui et al., 2017; Kleemans and Magruder, 2018; Corbi et al., 2021).

**DNWR** We investigate the extent to which wage rigidity plays a role in our results in two different ways. First, we calculate the municipality-level Kaitz index as the ratio between the national minimum wage and municipalities’ median wages (computed pooling formal and informal wage workers) for all municipalities in Brazil in 2000. We define as high DNWR a municipality with above median Kaitz index. Second, we borrow from Rodríguez-Clare et al. (2024) who measure DNWR using the share of workers with negative year-on-year hourly wage changes among workers with non-zero wage changes.<sup>17</sup> We compute in RAIS data the average share of year-on-year changes in 1997-98, 1998-99, and 1999-2000 at the micro-region level, and define as high DNWR a micro-region with below median share of negative wage changes.<sup>18</sup>

For both measures of formal wage rigidity, we estimate the following variation of regression 2:

$$\Delta y_d = \beta_0 + \beta_1 Mig_d + \beta_2 Mig_d \times D_d + \beta_3' \mathbf{X}_d + \varepsilon_d \quad (4)$$

where  $D_d$  is a wage rigidity dummy that equals one if municipality  $d$  has a high level of DNWR. We instrument  $Mig_d$  and  $Mig_d \times D_d$  with  $Z_d$  and  $Z_d \times D_d$ . All regressions are weighted by population in 2000.

Table C.1 presents the results. Panel A Columns 2 and 3 suggest that the positive effects of immigration on formal employment are more muted in municipalities with higher DNWR: Column 2 even indicates that there is no increase in formal employment when minimum wages are more binding. Panel B Columns 2 and 3 suggest that when formal wages are more rigid, migration increases unemployment. In addition, Panel A Column 5 suggest that informality decreases less when minimum wages are more binding, and Panel B Column 6 that formal wages decrease less when wages are more rigid. Taken together, these findings echo Harris and Todaro (1970) and Fields (1975)’s predictions that when formal wages are rigid downward rural-urban migration fosters urban unemployment and informality.

<sup>17</sup>We also compute this share among all workers and find identical results (not reported here).

<sup>18</sup>Averaging reduces noise from very small municipalities with a few formal workers. A micro-region is larger than a municipality, it corresponds to Commuting Zones in the US.

**Short-run Effects** To test whether differences in time horizon can indeed explain the discrepancy between our results and those found in the previous literature, we lean on the work by [Corbi et al. \(2021\)](#), who show that immigration from the semi-arid regions of Brazil increases informality and unemployment in urban destinations. We use the same data as theirs for destination outcomes, but we expand the sample to all sending municipalities (See Section 2).<sup>19</sup> Following [Corbi et al. \(2021\)](#) and the rest of the literature, we estimate the following year-on-year specification:

$$\Delta y_{dt} = \beta_0 + \gamma_1 Mig_{dt} + \gamma_2' \mathbf{X}_d + \eta_t + \varepsilon_d \quad (5)$$

where  $d$  denotes destination municipalities,  $\eta_t$  denote year fixed effects, and  $\Delta y_{dt} = y_{d,t} - y_{d,t-1}$  for  $t = 2001, \dots, 2010$ . The vector  $\mathbf{X}_d$  is the same as before and includes the share of female, young and high-skilled workers in municipality  $d$  in 2000.  $Mig_{dt}$  measures the inflow of migrants in year  $t$  to destination municipality  $d$ , which we compute using the Census. We instrument  $Mig_{dt}$  with a higher frequency version of the drought instrument (see Appendix C.2 for more details).

Table C.2 shows that municipalities that receive more migrants experience a reduction in formal wage employment, while informal employment and non-employment remain unaffected. The share of informal jobs is thus increasing in regions that receive more migrants. We find no statistically significant effects on formal wages. These striking differences between short- and long-run results are not driven by the restricted sample of 700 municipalities we use in the short-run specification. We obtain the same long-run results if we restrict our decadal change specification to the same sample of 700 municipalities available in the PNAD data (see Appendix Tables C.3 and C.4). Hence, our year-on-year analysis confirms that immigration leads to an increase in urban informality in the short run ([Kleemans and Magruder, 2018](#); [El Badaoui et al., 2017](#); [Corbi et al., 2021](#)).

The contrast between the short- and long-run results, together with the heterogeneity in effects between municipalities with more versus less DNWR, suggests that downward wage rigidity in the formal sector is an important determinant of the economic effects of rural-urban migration on urban labor markets. In the next sections, we develop and estimate a model of firm dynamics and informality which we use

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<sup>19</sup>We are grateful to the authors for providing us with the codes that identify municipalities in the National Household Survey, without which it would have not been possible to conduct this analysis.

to quantitatively assess the nexus between formal wage rigidity, short- and long-run labor market effects of immigration. We also perform counterfactual simulations to study how the informal sector shapes the effects of migration at destination.

## 4 Model

This section develops a simple equilibrium model of firm dynamics and informality. We build on [Hopenhayn \(1992\)](#), with key innovations in the entry structure, firms’ productivity processes and, crucially, the two margins of informality ([Ulyssea, 2018](#)). Importantly, this is not a spatial model that aims at capturing the equilibrium effects of migration for the country as a whole. Instead, we model a single urban economy, calibrate it to the average urban destination in Brazil, and use it to assess the combined equilibrium effects of the labor supply and demand shocks represented by immigration flows in the average destination economy, as well as to gain a deeper understanding of the mechanisms behind our empirical results.

### 4.1 Set up

Every period, there is a continuum of firms that are indexed by their idiosyncratic productivity,  $\theta$ . All firms have the same technology and use labor as their only input:  $f(\ell) = \theta q(\ell)$ , where  $q(\cdot)$  is increasing and concave. Formal and informal firms operate in the same industry, produce a homogeneous good and face the same prices in a competitive market. For simplicity, we assume that labour is homogeneous, which implies that formal and informal workers earn the same wage in steady state equilibrium.<sup>20</sup> The informal sector is composed by informal firms (the extensive margin), while informal workers can be found in both sectors (because of the intensive margin).

These assumptions regarding market structure imply complete integration between the formal and informal sectors, which is in sharp contrast with the “dual economy” view that goes back to [Lewis \(1954\)](#), [Harris and Todaro \(1970\)](#) and [Fields \(1975\)](#).

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<sup>20</sup>The model can accommodate different skill levels and imperfect substitution between migrants and non-migrants. Nevertheless, our empirical results show that: immigration was neutral in terms of the skill composition at destination; conditional on skills, internal migrants tend to be close substitutes to non-migrants; and effects are the same for skilled and non-skilled workers. Thus, we believe that the worker homogeneity assumption is a reasonable simplification in our context.

This view, however, is at odds with the more recent evidence available (e.g. [Hsieh and Olken, 2014](#)), in particular that formal and informal firms coexist within narrowly defined industries, there is a large degree of overlap between formal and informal firms' productivity distributions, and a large fraction of informal employment is in fact located within formal firms ([Ulyssea, 2020](#)).

## 4.2 Profits

The static, within-period problem follows very closely [Ulyssea \(2018\)](#). Informal firms are able to avoid taxes, but face an “informality cost” captured by  $1 \leq \tau_i(\cdot) < \infty$ , where  $\tau'_i, \tau''_i > 0$ . This function is a general formulation for the different costs associated to informality, such as the probability of being inspected by the government. The latter is likely to be increasing in firms' size, as larger firms are more visible to the government and inspected with higher probability, for example. Informal firms' profit function is given by

$$\Pi_i(\theta, w) = \max_{\ell} \{\theta q(\ell) - \tau_i(\ell) w\} \quad (6)$$

Formal firms must pay revenue and payroll taxes, but can evade the latter by hiring informal workers. However, formal firms also face a cost of hiring informal workers that can be rationalized along similar lines as above – e.g. a large formal firm is more likely to be audited by the government. We assume that formal firms' cost of hiring informal workers is given by  $1 \leq \tau_f(\ell_i) < \infty$ , where also  $\tau'_f, \tau''_f > 0$ . Hence, formal and informal workers have different marginal costs due to regulations that are imperfectly enforced. As both types of workers are perfect substitutes, at the margin firms hire the cheaper one and there is a unique threshold,  $\tilde{\ell}$ , above which all additional workers are hired formally.<sup>21</sup> Formal firms' profit function can thus be written as follows:

$$\Pi_f(\theta, w) = \max_{\ell} \{(1 - \tau_y) \theta q(\ell) - C(\ell)\} \quad (7)$$

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<sup>21</sup>The marginal cost of hiring informal workers is increasing and given by  $\tau'_f(\ell)w$ , while the marginal cost of hiring formal workers is constant and given by  $(1 + \tau_w)w$ , where  $\tau_w$  is the payroll tax. The threshold  $\tilde{\ell}$  is given by the point where these two marginal costs are equated.

where  $\tau_y$  denotes the revenue tax and

$$C(\ell) = \begin{cases} \tau_f(\ell) w, & \text{for } \ell \leq \tilde{\ell} \\ \tau_f(\tilde{\ell}) w + (1 + \tau_w)(\ell - \tilde{\ell}) w, & \text{for } \ell > \tilde{\ell} \end{cases} \quad (8)$$

Additionally, every period firms must pay a fixed cost of operation and the net profit is given by:  $\pi_s(\theta, w) = \Pi_s(\theta, w) - \bar{c}_s$ ,  $s = i, f$ .

### 4.3 Entry and Dynamics

Every period, there is a large mass of potential entrants that decide whether to enter the informal or formal sectors, or not to enter at all. Before entry, they observe their potential long-run (fixed) productivity parameter,  $\nu$ , which is drawn from the CDF  $H(\nu)$ . These are assumed to be i.i.d., such that entry in one period does not affect the composition of entrants in the following period. Entrants in both sectors must pay a fixed entry cost,  $c_s^e$ . These parameters will be estimated, but we expect that  $c_f^e > c_i^e$  due to regulatory costs associated with formal firm creation.

Post-entry dynamics are driven by the evolution of firms' idiosyncratic productivity,  $\theta$ , which is the only source of uncertainty to the firm (there are no aggregate shocks). The productivity process is characterized by the following expressions:

$$\ln \theta_{j,1} = \ln \nu_j + \ln \epsilon_{j,1} \quad (9)$$

$$\ln \theta_{j,t} = \rho_s \ln \theta_{j,t-1} + (1 - \rho_s) \ln \nu_j + \ln \epsilon_{j,t}, \quad t \geq 2 \quad (10)$$

where  $j$  indexes firms, the unexpected shock is i.i.d. and  $\ln \epsilon \sim \ln \mathcal{N}(0, \sigma_s^2)$ ,  $s = i, f$ .

Equation 10 is very close to standard formulations in the dynamic panel literature (e.g. [Blundell and Bond, 1998](#)), where firms' long-run productivity level is given by  $\ln \theta_\infty = \ln \nu$ . Crucially, Equations 9 and 10 imply that firms' current productivity can differ from their long-run productivity level, and that there is heterogeneity across firms in terms of life-cycle growth profiles.

We assume that formal firms cannot become informal,<sup>22</sup> and therefore face a simple

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<sup>22</sup>Once the firm is visible to the government, it is very hard to become informal and therefore “invisible” again, as the government has formal firms' complete information. Firms could change locations to become informal, which in our model would correspond to exiting the formal sector and rejoining the pool of potential entrants (but with no memory of previous experiences).

stopping-time problem (to remain active or exit). Informal incumbents have the additional option of transiting to the formal sector. To formalize, an informal firm must pay the difference between entry costs,  $\tilde{c}^e = c_f^e - c_i^e$ , and after that it will face the regulatory costs implied by formality. In addition to endogenous exit, we assume that formal and informal firms face exogenous death shocks denoted by  $\delta_s$ .

The value functions of formal and informal *incumbents*, respectively, can be written in recursive form as follows:

$$V_f(\theta, w) = \pi_f(\theta, w) + (1 - \delta_f) \beta \max \{0, E_\nu [V_f(\theta', w) | \theta]\} \quad (11)$$

$$\begin{aligned} V_i(\theta, w) &= \pi_i(\theta, w) \\ &+ \beta \max \{0, (1 - \delta_i) E_\nu [V_i(\theta', w) | \theta], (1 - \delta_f) E_\nu [V_f(\theta', w) | \theta] - \tilde{c}^e\} \end{aligned} \quad (12)$$

where  $\beta$  denotes the discount factor and the subindex  $\nu$  in the continuation value indicates that the expectation depends on the firm-specific long-run productivity parameter, as described in expressions 9 and 10.

The exit decision is made before future productivity is revealed and it follows a cut-off rule. If  $\theta < \underline{\theta}_s$  firms choose to exit, where  $\underline{\theta}_s$  is given by

$$E_\nu [V_s(\theta', w) | \underline{\theta}_s] = 0, \quad s = i, f \quad (13)$$

Entry is also characterized by a threshold rule, as follows:

$$E_\nu [V_i(\theta, w) | \nu = \underline{\nu}_i] = c_i^e \quad (14)$$

$$E_\nu [V_f(\theta, w) - V_i(\theta, w) | \nu = \underline{\nu}_f] = c_f^e - c_i^e \quad (15)$$

where  $\underline{\nu}_s$  characterizes the last firm to enter sector  $s = i, f$ .

Finally, informal firms formalize if  $\theta \geq \bar{\theta}_i$ , where  $\bar{\theta}_i$  is given by:

$$E_\nu [V_f(\theta', w) - V_i(\theta', w) | \bar{\theta}_i] = \tilde{c}^e \quad (16)$$

The *timing* of informal firms' decisions is the following. At the beginning of the period, firms draw their productivity shock,  $\theta$ , and decide how much to produce (i.e. how much labor to hire). If  $\theta \in [\underline{\theta}_i, \bar{\theta}_i)$ , they will start next period in the informal sector again; if  $\theta \geq \bar{\theta}_i$ , they will pay  $\tilde{c}^e$  and start next period in the formal sector; and if  $\theta < \underline{\theta}_i$ , they exit. The timing for formal firms is the same, except that they

only choose whether to stay or exit.

## 4.4 Steady state equilibrium

To close the model, we assume that there is a representative household that inelastically supplies  $\bar{L}$  units of labor and that derives utility from consuming the final good. We model migration as a once-and-for-all increase in  $\bar{L}$ .<sup>23</sup> Households consume all of their income, which is given by  $w\bar{L} + \Pi + T$ , where  $\Pi$  denotes the mass of profits in the economy,  $w\bar{L}$  is total wage income and  $T$  denotes total tax revenues. Therefore, migration simultaneously increases labor supply and demand for the final good.

We focus on stationary competitive equilibria, which correspond to a set of allocations, wage, cutoffs and measures of firms such that they remain constant over time and the following conditions hold in every period: (i) product and labor markets clear;<sup>24</sup> (ii) the cutoffs  $(\underline{\theta}_s, \bar{\theta}_s, \underline{\nu}_s)$ ,  $s = i, f$ , are determined according to (13)–(16) and entry conditions hold. In Appendix D we show that there exists a unique stationary competitive equilibrium where the formal and informal sectors exist and have positive entry and exit (Proposition 1).

Appendix D provides other characterization results that shed light on the model's main mechanisms. In particular, we show that the productivity distribution among active firms within a sector and cohort is increasing in the cohort's age (Proposition 2). Put differently, the productivity distribution of older cohorts first-order stochastically dominates that of younger cohorts. From this result, it follows that any function that is increasing in firms' productivity,  $\theta$ , will also be increasing in cohort's age. In particular, this is true for the survival rate, firms' average size and average revenue in *both the formal and informal sectors* (Corollary 1). These results also imply that the average share of informal workers within formal firms (intensive margin) and the average share of informal firms (extensive margin) within a given cohort decline with the cohort's age (Corollaries 2 and 3, respectively). Importantly, these results constitute falsifiable predictions and show that the model has empirical content.

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<sup>23</sup>Our empirical results show that immigration has no effect on unemployment nor on the share of individuals out of the labor force. Hence, we do not model unemployment nor labor force participation decisions.

<sup>24</sup>We normalize the price of the final good to one throughout the analysis.

## 4.5 Discussion

An immigration shock brings about an increase in both labor supply and demand for the final good at destination. The higher labor supply would lead to a decline in real wages, which combined with the higher demand for the final good would cause incumbents' labor demand to expand. However, incumbents are eventually limited by scale, so the increase in labor supply is likely to be also met by greater firm entry. In equilibrium, not only entry rates are affected, but also exit rates, and firm size distribution. The literature that explains the “startup deficit” in the US by a slowdown in labor supply growth highlights the same mechanisms working in reverse ([Hopenhayn et al., 2018](#); [Karahan et al., 2019](#)).

It is unclear, however, what the net effects should be in terms of overall firm and labor informality, or output per worker. If migrants are largely absorbed by informal jobs in informal firms or low-productivity formal firms, the net result will most likely be an increase in informality among firms and workers. However, if the decline in wages induces greater entry also of firms with higher growth potential – either in the informal or formal sectors – then this can lead to more firm growth in the formal sector, and to a new steady state equilibrium with lower informality. The results discussed in [Section 3](#) suggest that the latter force dominates, but these are relative effects between municipalities more and less exposed to migration. We now move to the model calibration and counterfactuals to quantify these effects in equilibrium.

## 5 Counterfactuals

### 5.1 Model Calibration

To calibrate the model discussed in [Section 4](#), and use it to perform counterfactual simulations, we must parameterize the model's remaining objects that were left unspecified. Starting by the production function, we assume a simple span-of-control formulation:  $y(\theta, \ell) = \theta \ell^\alpha$ ,  $0 < \alpha < 1$ . The cost functions are defined as in [Ulyssea \(2018\)](#):  $\tau_s(\ell) = \left(1 + \frac{\ell}{\varphi_s}\right) \ell$ , where  $\varphi_s > 0$  and  $\tau'_s, \tau''_s > 0$ . The larger  $\varphi_i$  the more informal firms can grow before it becomes too costly to be informal, and the larger  $\varphi_f$  the easier it is for formal firms to hire informal workers. We assume that the potential long-run productivity parameter,  $\nu$ , follows a Pareto distribution. Hence,



firms' first productivity draw,  $\theta_1 = \nu\epsilon_1$ , has a Pareto-Lognormal distribution. This distribution is very well-suited to characterize firm size distributions in developing countries.

We use a two-step minimum distance calibration. In the first step, we use the RAIS establishment-level panel to estimate the persistence parameter of formal firms' productivity process, which gives  $\rho_f = 0.921$  (see Appendix E for details). We set the tax parameters to their statutory values:  $\tau_w = 0.375$  and  $\tau_y = 0.293$ . The value of  $\tau_w$  corresponds to the main regulatory costs that are proportional to firms' payroll (social security contribution, direct payroll tax and severance contributions). The  $\tau_y$  corresponds to the federal VAT taxes (IPI and PIS/COFINS), and excludes state-level value-added taxes, which vary across state borders and create an intricate system of tax substitution along the production chain. As in Ulyssea (2018), the Pareto distribution scale parameter ( $\nu_0$ ) is set so that the firms' minimum size is one employee, while formal sector's fixed cost of operation parameter is set to be 70 percent of the baseline wage. Table 4 describes all parameter values.

The second step takes the first-step parameters as given to calibrate the remaining 12 parameters of the model, which are the following:

$$\Omega = \{\varphi_f, \varphi_i, \delta_i, \delta_f, \bar{c}_i, \xi, c_f^e, c_i^e, \alpha, \sigma_i, \sigma_f, \rho_i\}$$

where  $\varphi_i$  and  $\varphi_f$  are the parameters of extensive and intensive margins' cost functions, respectively;  $\delta_i$  and  $\delta_f$  are the exogenous death shocks in the informal and formal sectors, respectively;  $\bar{c}_i$  determines the per-period fixed cost of operation;  $\xi$  is the Pareto shape parameter;  $c_f^e$  and  $c_i^e$  are the formal and informal entry costs, respectively;  $\alpha$  is the span-of-control;  $\sigma_i^2$  and  $\sigma_f^2$  are the informal and formal variances of the productivity shock; and  $\rho_i$  is the persistence of the productivity process in the informal sector. The vector  $\Omega$  is estimated by minimizing the distance between moments from the data,  $\hat{m}_N$ , and moments computed from simulated data generated by the structural model. Appendix Section E provides more details about the implementation.

## Moments and parameters

We use two main data sources to estimate the moments used in the calibration: the RAIS, which covers all formal firms, and the ECINF survey which is representative of all formal and informal firms with up to five employees (See Section 2). Since the

ECINF survey is only available in 2003, we restrict the RAIS to the same year when computing static moments. For moments relating to firm growth, we use a 2000-2011 panel from the RAIS, the same period used in the IV analysis. We compute the following moments: overall share of informal firms and by size brackets (less than 2, and 3 to 5 employees); average share of informal workers within formal firms; formal and informal firm growth at ages 5 and 10 relative to age 1; formal firms size distribution by size brackets (less than 5, 5 to 10, 11 to 20, 20 to 50, and more than 50 employees); informal firms size distribution by size brackets (less than 2, and 3 to 5 employees). We compute the share of informal workers in the economy using the 2003 National Household Survey (PNAD).

Table 4 shows the results. The estimated cost of entry in the formal sector is more than twice as large as for the informal sector. The values denominated in 2003 *Reais* are not negligible: formal sector's entry costs correspond to more than 30 times the monthly national minimum wage at the time. Informal firms face a relatively high exogenous death shock of nearly 15 percent ( $\delta_i = 0.148$ ), which is more than twice as large as in the formal sector. Apart from the exogenous exit shock, informal firms do not face higher uncertainty, as the variances of the productivity innovation in both sectors,  $\sigma_s^2$ , are very similar. The same is true for persistence of the productivity process in both sectors.

## Model fit

Table 5 shows the main targeted data moments and their model counterparts. To further assess how the model fits the data, we examine non-targeted moments in Figure 4. Panels (a) and (b) show that the model reproduces well the behavior observed for the extensive and intensive margins of informality, respectively. The model predicts a steeper gradient in the share of informal firms by firm size than what is observed in the data, but the fit for the intensive margin is very good and the model is able to match well the behavior of the average share of informal workers within formal firms relative to firm size.

We also use the panel structure of RAIS to examine how the model fits moments directly related to formal firms' dynamics. As panel (c) shows, the model fits well the growth profile for formal firms up to age 11, which is the maximum age that we are able to measure in our panel (see Appendix). The simulated data shows a

more concave profile, however, which implies that the model slightly overestimates the accumulated growth for ages 3 to 9, but then converges to the data at age 11. Similarly, panel (d) in Figure 4 shows that the model reproduces very well the behavior of the autocorrelations of log-employment relative to log-employment at age one. This is reassuring, as these moments are never used in the estimation, and are quite informative about the structure of the firms’ productivity process, in particular the importance of permanent components (Sterk et al., 2021).

In the Appendix E, we provide a careful discussion of which dimensions of the data provide the variation that allows us to pin down the different model parameters and investigate how informative the moments used in the calibration are.

## 5.2 Results

We start our quantitative analysis by investigating the equilibrium effects of a large immigration shock. We simulate a once-and-for-all increase in the aggregate labor supply of 10 percent, which is around the 80th percentile of the distribution of immigration rates we observe in our data. To compare our simulation and empirical results, we run a modified version of the IV specification described in Section 3.1 (Expression 2). We replace the immigration rate,  $Mig_d$ , with a dummy that equals one if the municipality is above the median immigration rate and zero otherwise, and instrument it using the same shift-share instrument based on drought shocks. In the data, the average migration rate below median is 3.2 percent, and the average above median is 10.9 percent. Hence our specification allows us to estimate the average effect of receiving a large immigration shock relative to receiving fewer migrants, which is comparable to our first counterfactual.

Table 6 present the effects of the immigration shock according to the IV estimates (Column 1) and to the counterfactual simulation (Column 2). The model fits the empirical results well for the three outcomes that are directly comparable. The counterfactual simulation shows a 4.3 percent decline in the share of informal workers, while the IV estimation shows a 7.3 percent decline. The effects on the number of formal firms are also comparable, with model simulation and IV estimation implying an increase of 16.3 and 20.7 percent, respectively. The model shows a 3.4 percent wage reduction compared to 5.3 in the IV estimation, thus underestimating the overall wage decline. Given the model’s simplifying assumptions – in particular the lack of

worker heterogeneity – it is not surprising that it underestimates the wage variation, albeit it remains close to the reduced-form effects. Given that the wage decline is smaller than the IV estimate, it is natural that the model simulated increase in the number of formal firms and formalization effects are also smaller than their empirical counterparts.

The model allows us to go further and assess the source of this increase in the number of formal firms. As Column 2 in Table 6 shows, around 40 percent comes from the formalization of previously existing informal firms, while the remaining 60 percent comes from the creation of new formal businesses. This result highlights the importance of accounting for firm dynamics and the linkages between the informal and formal sectors. A static framework would largely underestimate the increase in the number of formal firms, as it would not be able to account for the transition of informal firms to the formal sector after entry occurs. Similarly, the dual view of informality – which rules out such transitions by assumption – would miss this important dimension of the economy’s adjustment to shocks. This result also suggests that the informal sector might play a positive role in the adjustment process of economies exposed to labor supply shocks.

Column 2 in Table 6 also shows the equilibrium effects on additional key aggregate outcomes. First, immigration leads to a 5.3 percent decline in the share of informal firms in the economy (from 69.6 to 65.9 percent), which confirms that the increase in the number of active firms in the economy (of 3.7 percent) is driven by the increase in the number of formal firms. Second, even though informality is declining, average firm productivity in the economy declines by 1.4 percent. As Panel (a) in Figure 5 shows, this is the result of worsening firm composition within the formal sector: the share of firms in the lower productivity quartile increases by more than 4 percentage points, while the share in the upper quartile declines by around 3 percentage points. Employment composition changes in a similar but less pronounced way. Panel (b) in Figure 5 also documents a very small positive effect on formal firm growth.

Finally, output and taxes increase by 7.1 and 8.7 percent respectively, and therefore decline in per capita terms. This is not surprising given the results on firm composition. Nevertheless, it is worth highlighting that these results are likely to be lower bounds of the effects on productivity and output, as we assume that there are no technological differences between the formal and informal sectors, nor differential access to capital. If firms that formalize gain access to cheaper credit or better tech-

nologies, then one could expect effects on output and average firm productivity to become more positive.

### From Short- to Long-Run: Wage Rigidity in the Formal Sector

Following the discussion in Section 3.5, we now use the model to quantitatively assess the role of downward wage rigidity in the formal sector during the transition between steady states after the economy is hit with an immigration shock. Although formal wages take time to adjust due to DNWR, we assume that wages in the informal sector are fully flexible in every period.

In period  $t = 0$  the economy is in the initial steady state, and the immigration shock hits at  $t = 1$ . We treat this as a “MIT shock”, which is an unpredictable shock to the steady state equilibrium of an economy without aggregate shocks (Boppart et al., 2018). We focus on the equilibrium along a perfect-foresight path, and under the assumption that no shock will ever hit this economy again. In the final period,  $t = T$ , the economy reaches the new steady state equilibrium discussed in above (Column 2 in Table 6), wages in the formal sector have fully adjusted, and wages in both sectors are equalized. We follow Schmitt-Grohé and Uribe (2016) and assume that during the transition between steady states wages in the formal sector evolve as follows:  $w_{f,t} = \gamma w_{f,t-1}$ , where we set  $\gamma = 0.996$  so that it takes 10 years for formal wages to reach the new steady state value (see Appendix Section F for more details).

Figure 6 shows the results for the evolution of the share of informal workers and firms. In the immediate short run, both labor and firm informality increase substantially, as the formal sector wage cannot adjust and the labor supply shock is entirely absorbed by informal jobs. The number of informal firms shows limited increase in the first two years (around 1.8 percent), and the employment adjustment largely occurs via the intensive margin of informality. This also explains why the share of informal labor hovers around its initial level for a few years before falling, while firm informality starts declining almost immediately after the shock: the new formal firms tend to be small and lower productivity, and therefore hire a substantial fraction of their labor force informally. Finally, the economy performs worse during the transition when compared to the final steady state when wages have fully adjusted: output, average firm productivity, and tax revenues are substantially lower.

## Immigration shock without the informal sector

Results in Column 2 in Table 6 suggest that a substantial part of the increase in the number of formal firms comes from informal firms that formalize. The results relative to the transition dynamics also show that informality plays an important role in the short-run absorption of the increased labor force following an immigration shock. The informal sector therefore acts as a “stepping stone” for entry into the formal sector, which could lead to better long-run aggregate effects of immigration relative to a counterfactual with stricter enforcement and lower informality. We directly investigate this conjecture by combining the same labor supply shock with nearly shutting down the extensive margin of informality. This corresponds to making the cost function of operating in the informal sector very steep by substantially reducing the parameter  $\varphi_i$ . Conceptually, this simulates a scenario in which the government devotes a lot of resources to increasing enforcement on informal firms, making it prohibitively costly for firms to operate unregistered, unless they are very small. Given the costs of such a policy, we see this counterfactual as a way to analyze the role of the informal sector rather than as a feasible policy experiment.

As Column 3 in Table 6 shows, nearly shutting down the extensive margin of informality leads to a substantial reduction in the share of informal firms and workers relative to the equilibrium with the labor supply shock alone (second column). This comes at the cost of a sizable reduction in the total number of firms in the economy, as many low productivity informal firms do not survive. As a result, the effect on firm composition is positive and average firm productivity increases by 2.5 percent, as opposed to a reduction of 1.4 percent with the labor shock without changes in enforcement. Consistently, the effect on total output is higher than in the basic labor supply shock scenario (8.3 and 7.1 percent, respectively), and the effect on taxes is much higher, as the economy is now much more formal than in the baseline.

Overall, the economy reaps larger dividends from the labor supply shock when the informal sector is nearly shut down. Thus, the informal sector is dampening the economy’s ability to reap greater dividends of a growing labor force. This echoes the results from [Dix-Carneiro et al. \(2024\)](#), who show that even though the informal sector acts as an “employment buffer” when the economy faces a negative shock, it is not a “welfare buffer” since the counterfactual with stricter enforcement still leads to a higher welfare.

## 6 Conclusion

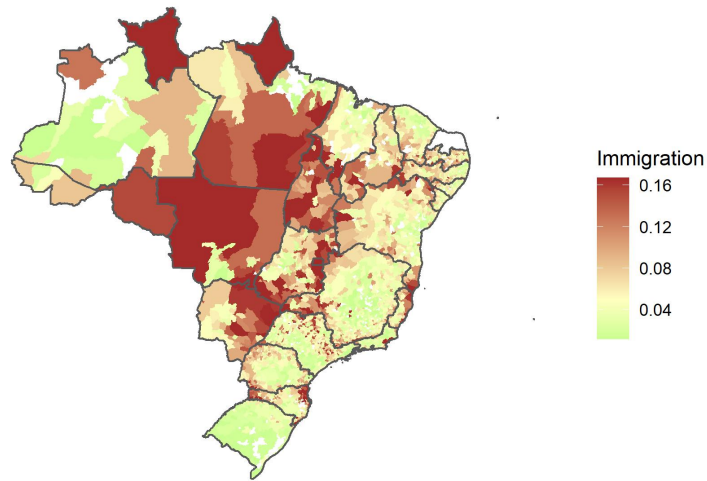
We study the economic effects of rural-urban migration on Brazilian cities between 2000 and 2010 using a shift-share IV design. We show that immigration fosters formal employment and formal firm creation while lowering formal and informal sector wages, with no effect on unemployment. These results run counter to the view that rural-urban migration increases the number of informal or unemployed workers in developing country cities, a view that dates back to the theoretical arguments of [Harris and Todaro \(1970\)](#) and [Fields \(1975\)](#) and has been confirmed by recent empirical evidence ([Kleemans and Magruder, 2018](#); [Corbi et al., 2021](#)).

To rationalize these surprising results, we develop and calibrate a structural model of firm dynamics and informality. The model quantitatively replicates the long-run IV results: large labor supply shocks generate wage reductions, and substantial gains in formal employment and formal firm creation. If we impose sluggish wage adjustment in the formal sector during the transition between steady states, we find that labor and firm informality increase in the short-run, but converge to the new lower levels in the longer run. Despite its role as a “stepping-stone” in the short run, the informal sector reduces the aggregate benefits of immigration.

Our results suggest that, in the long run, urban developing economies might experience demographic dividends from internal climate migration ([Rigaud et al., 2018](#)).

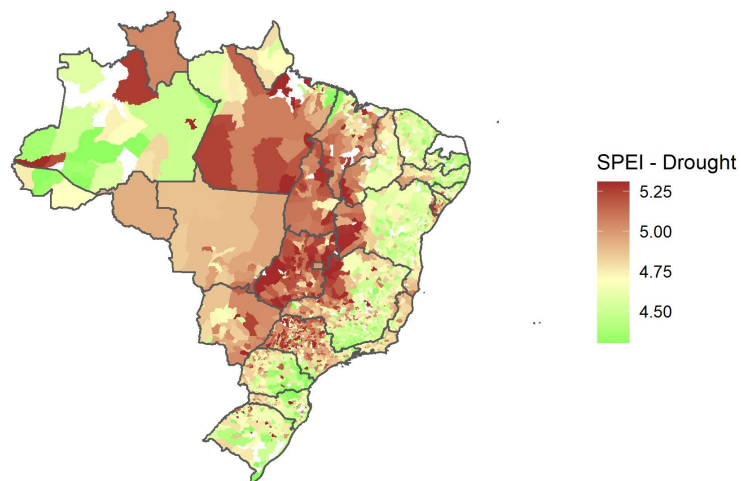
## Tables and Figures

Figure 1: In-Migration, 2000-2010



Notes: Computed using the Decennial Population Census. Darker areas denote higher in-migration rates.

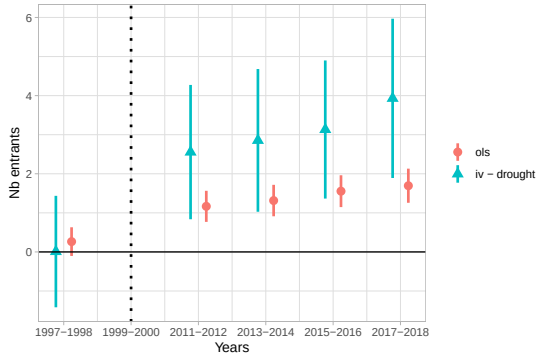
Figure 2: Drought Shocks



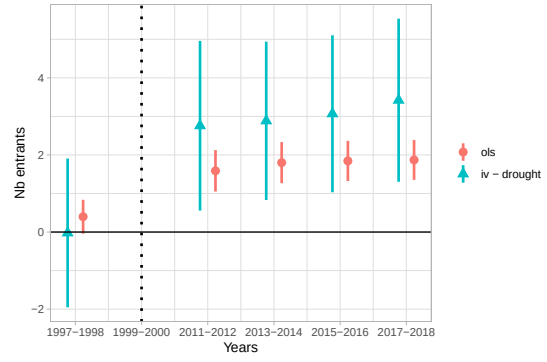
Notes: The Drought shocks are constructed according to expression 1 in the text.



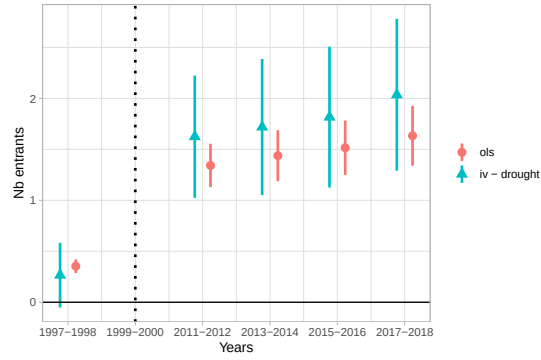
Figure 3: Pre-Trends and Effects Beyond 2011-12



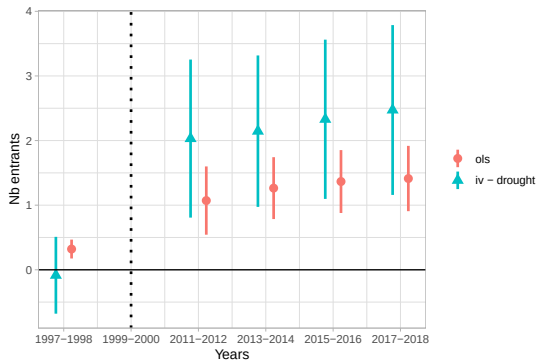
(a) Number of entrants



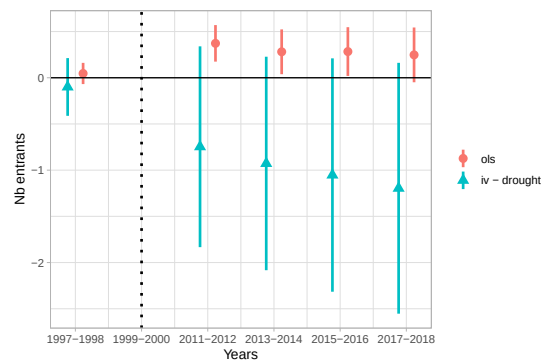
(b) Number of exiting firms



(c) Number of firms

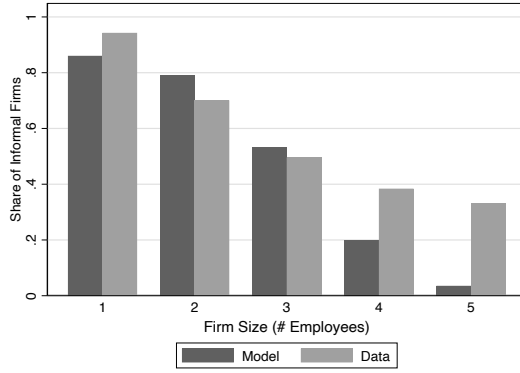


(d) Number of jobs

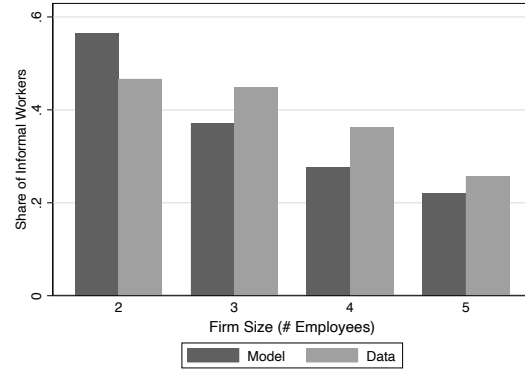


(e) Firm wage

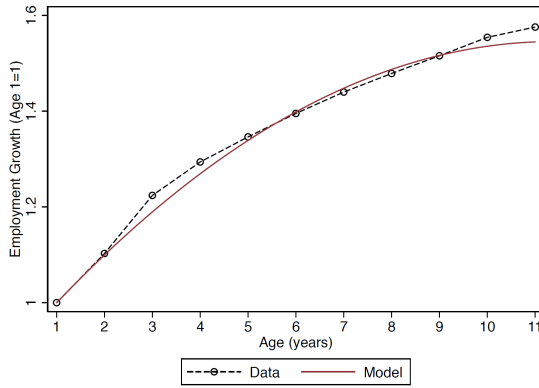
Figure 4: Model fit: non-targeted moments



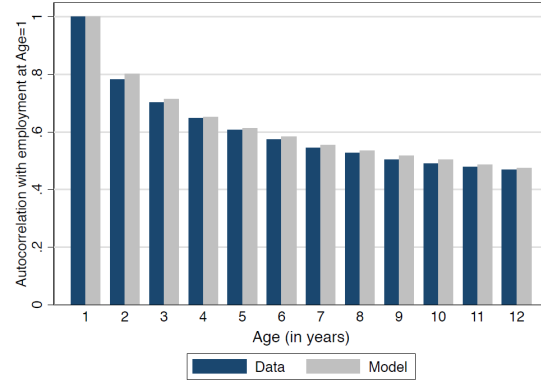
(a) Extensive Mg.



(b) Intensive Mg.



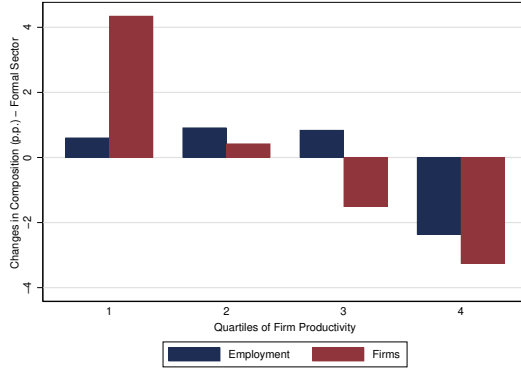
(c) Formal Firms' growth



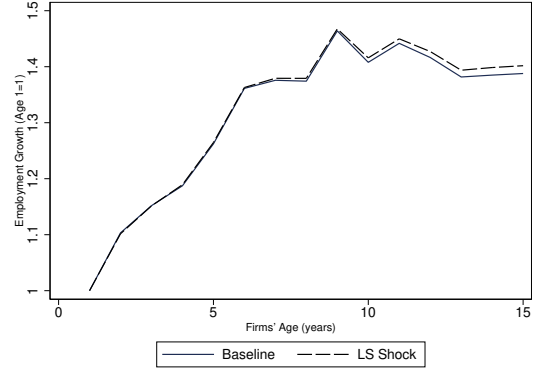
(d) Log-Employment Autocorrelations

Notes: Size is measured as number of employees. For panels (c) and (d), both simulated and empirical moments use only formal firms and their formal workers. The empirical moments are computed using the panel data from RAIS.

Figure 5: Counterfactuals: Formal Sector Firms

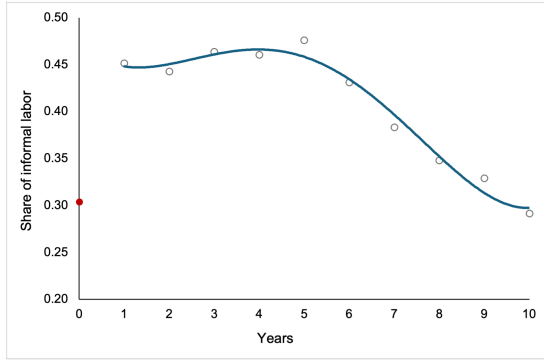


(a) Changes in Composition

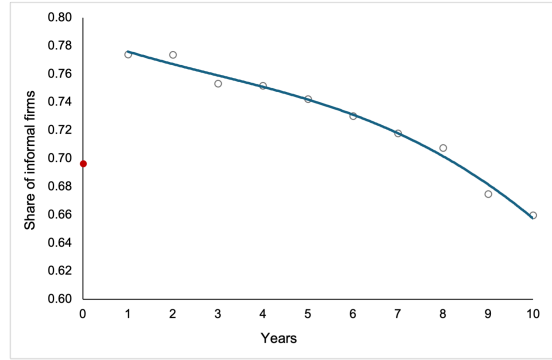


(b) Growth Profiles

Figure 6: Transition Dynamics – Informality



(a) Labor Informality



(b) Firm Informality

Notes: Red dot indicates the initial steady state of the baseline economy.

Table 1: Descriptive Statistics

|                                   | 2000   |        | 2010   |        |
|-----------------------------------|--------|--------|--------|--------|
|                                   | Mean   | SD     | Mean   | SD     |
| Panel A: Socio-Demographics       |        |        |        |        |
| % Female                          | 0.483  | 0.014  | 0.482  | 0.013  |
| % Young                           | 0.128  | 0.021  | 0.102  | 0.016  |
| % Low-skilled                     | 0.72   | 0.096  | 0.579  | 0.094  |
| Panel B: Labor Market Outcomes    |        |        |        |        |
| % Overall wage employment         | 0.332  | 0.061  | 0.4    | 0.071  |
| % Formal wage employment          | 0.229  | 0.076  | 0.309  | 0.096  |
| % Informal wage employment        | 0.103  | 0.036  | 0.09   | 0.037  |
| Log overall monthly wage          | 6.886  | 0.415  | 6.992  | 0.321  |
| Log formal monthly wage           | 7.017  | 0.361  | 7.074  | 0.28   |
| Log informal monthly wage         | 6.578  | 0.391  | 6.712  | 0.296  |
| Panel C: Firm Outcomes            |        |        |        |        |
| Log number of firms               | 8.728  | 2.341  | 9.286  | 2.195  |
| Log entry                         | 7.169  | 2.273  | 7.58   | 2.117  |
| Log exit                          | 6.918  | 2.345  | 7.306  | 2.14   |
| Log number of jobs                | 11.303 | 2.514  | 11.955 | 2.389  |
| Log firm monthly wage             | 7.091  | 0.455  | 7.376  | 0.322  |
| Panel D: Immigration              |        |        |        |        |
| Immigration rate                  |        |        | 0.176  | 0.092  |
| Immigration rate (State to state) |        |        | 0.07   | 0.054  |
| Panel E: Population               |        |        |        |        |
| Population                        | 25135  | 155267 | 31436  | 183082 |

Notes: For all variables there are 3,545 observations in both 2000 and 2010. Mean and Standard Deviation (SD) in panels A-D are weighted by population in 2000.

Table 2: Effects of Immigration on Workers

|                     | Wage employment   |                     |                      | Log monthly wage  |                     |                   |
|---------------------|-------------------|---------------------|----------------------|-------------------|---------------------|-------------------|
|                     | Overall<br>(1)    | Formal<br>(2)       | Informal<br>(3)      | Overall<br>(4)    | Formal<br>(5)       | Informal<br>(6)   |
| Panel A: OLS        |                   |                     |                      |                   |                     |                   |
| Immigration         | 0.037*<br>(0.019) | 0.105***<br>(0.023) | -0.068***<br>(0.014) | 0.062<br>(0.076)  | 0.032<br>(0.068)    | 0.035<br>(0.092)  |
| Panel B: IV-Drought |                   |                     |                      |                   |                     |                   |
| Immigration         | -0.014<br>(0.060) | 0.271***<br>(0.089) | -0.284***<br>(0.072) | -0.126<br>(0.284) | -0.671**<br>(0.336) | -0.200<br>(0.352) |
| F Statistic (IV)    | 18.11             | 18.11               | 18.11                | 18.11             | 18.11               | 18.11             |
| Baseline Mean       | 0.332             | 0.229               | 0.103                | -                 | -                   | -                 |
| Observations        | 3,545             | 3,545               | 3,545                | 3,545             | 3,545               | 3,545             |

Notes: Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000.

Table 3: Effects of Immigration on Firms

|                     | Nb firms<br>(1)     | Entry<br>(2)        | Exit<br>(3)         | Nb jobs<br>(4)      | Firm wage<br>(5)    |
|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Panel A: OLS        |                     |                     |                     |                     |                     |
| Immigration         | 1.342***<br>(0.108) | 1.167***<br>(0.203) | 1.588***<br>(0.274) | 1.071***<br>(0.270) | 0.372***<br>(0.101) |
| Panel B: IV-Drought |                     |                     |                     |                     |                     |
| Immigration         | 1.625***<br>(0.306) | 2.555***<br>(0.877) | 2.758**<br>(1.123)  | 2.031***<br>(0.624) | -0.747<br>(0.554)   |
| F Statistic (IV)    | 18.11               | 18.11               | 18.11               | 18.11               | 18.11               |
| Observations        | 3,545               | 3,545               | 3,545               | 3,545               | 3,545               |

Notes: Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000. Columns 1-3 refer to the log of total number of firms, entrants and exiting firms, respectively.

Table 4: Parameters of structural model

| Parameter                                  | Description                                                               | Source           | Value |
|--------------------------------------------|---------------------------------------------------------------------------|------------------|-------|
| <i>First Step</i>                          |                                                                           |                  |       |
| $\tau_w$                                   | Payroll Tax                                                               | Statutory values | 0.375 |
| $\tau_y$                                   | Revenue Tax                                                               | Statutory values | 0.293 |
| $\rho$                                     | Productivity Process: Persistence Parameter                               | GMM Estimation   | 0.92  |
| $\nu_0$                                    | Pareto's Location Parameter                                               | Calibrated       | 7.3   |
| $\gamma_f$                                 | Per-period fixed cost (Formal)                                            | Calibrated       | 0.7   |
| <i>Second Step: Min. Dist. Calibration</i> |                                                                           |                  |       |
| $\varphi_f$                                | Intensive margin: $\tau_f = \left(1 + \frac{\ell}{\varphi_f}\right) \ell$ | —                | 5.830 |
| $\varphi_i$                                | Extensive margin: $\tau_i = \left(1 + \frac{\ell}{\varphi_i}\right) \ell$ | —                | 5.427 |
| $\delta_i$                                 | Informal death shock                                                      | —                | 0.148 |
| $\delta_f$                                 | Formal death shock                                                        | —                | 0.066 |
| $\gamma_i$                                 | Per-period fixed cost (Informal)                                          | —                | 0.340 |
| $\xi$                                      | Pareto shape parameter                                                    | —                | 3.801 |
| $c_f^{e\dagger}$                           | Formal sector's entry cost                                                | —                | 6,205 |
| $c_i^{e\dagger}$                           | Informal sector's entry cost                                              | —                | 2,800 |
| $\alpha$                                   | Span-of-control                                                           | —                | 0.643 |
| $\sigma_i$                                 | Informal productivity process: SD                                         | —                | 0.144 |
| $\sigma_f$                                 | Formal productivity process: SD                                           | —                | 0.145 |
| $\rho_i$                                   | Informal productivity process: persistence                                | —                | 0.935 |

<sup>†</sup> Estimates and SD expressed in R\$ of 2003.

Table 5: Model Fit – Targeted moments

|                                  | Model | Data  |
|----------------------------------|-------|-------|
| Share Informal workers           | 0.304 | 0.301 |
| Share Informal Firms             | 0.696 | 0.696 |
| Informal Firms Size Distribution |       |       |
| ≤ 2 employees                    | 0.933 | 0.957 |
| ≤ 5 employees                    | 0.999 | 0.998 |
| Formal Firms Size Distribution   |       |       |
| ≤ 5 employees                    | 0.658 | 0.697 |
| 6 to 10                          | 0.136 | 0.144 |
| 11 to 20                         | 0.092 | 0.083 |
| 21 to 50                         | 0.053 | 0.048 |
| > 50                             | 0.023 | 0.028 |

Notes: Data moments computed using the RAIS, ECINF and PNAD data sets.

Table 6: Counterfactuals

|                                          | IV<br>Estimation | Labor Supply<br>Shock | LS Shock +<br>Enforcement |
|------------------------------------------|------------------|-----------------------|---------------------------|
| Share Informal Workers ( $\Delta\%$ )    | -7.3             | -4.3                  | -38.2                     |
| Wages ( $\Delta\%$ )                     | -5.3             | -3.4                  | -2.1                      |
| Number Formal Firms ( $\Delta\%$ )       | 20.7             | 16.3                  | 128                       |
| Newly created firms                      | —                | 9.9                   | —                         |
| Previously informal firms                | —                | 6.4                   | —                         |
| Share Informal Firms ( $\Delta\%$ )      | —                | -5.3                  | -68.2                     |
| Average Firm Productivity ( $\Delta\%$ ) | —                | -1.4                  | 2.5                       |
| Output ( $\Delta\%$ )                    | —                | 7.1                   | 8.3                       |
| Taxes ( $\Delta\%$ )                     | —                | 8.7                   | 30.9                      |

Notes: IV estimation results from a regression contrasting municipalities below and above the median immigration rate (see text). Labor supply shock corresponds to simulating a permanent increase of 10% in population, and effects are measured as percentage change relative to baseline values.

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# ONLINE APPENDIX

## A Additional results

### A.1 First Stage

We start this Section by reporting First stage results in Table A.1. As the table shows, both instruments are very predictive of immigration flows.

| Table A.1: First stage |                   |                  |
|------------------------|-------------------|------------------|
|                        | Immigration       |                  |
|                        | (1)               | (2)              |
| Price                  | −0.053<br>(0.006) |                  |
| Drought                |                   | 0.077<br>(0.004) |
| Observations           | 3545              | 3545             |

Notes: Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality’s population in 2000.

### A.2 Additional Results for Workers

In Table A.2 we use the benchmark IV specification to examine the impacts of immigration on occupational structure by looking at all possible employment statuses. Table A.3 reproduces the main results in Table 2, but computing labor market shares using total hours worked (columns 1–3), and log hourly wages for all, formal and informal workers (columns 4-6).

Table A.4 investigates the effects of immigration flows on labor force composition at destination using both IV-price and IV-droughts. We estimate the IV regression described in 2 without any controls, and having as dependent variables the share of females, low skill and young individuals in the population (columns 1–3), population change and out-migration rate, all computed at destination. Table A.5 shows the results estimated separately for high- and low-skilled workers, and Table A.6 separately for migrants and non-migrants.

Finally, Table A.7 investigates whether migrants are more likely to be business owners in municipalities that receive higher inflows of migrants. We do the same

Table A.2: Effects on Occupational Composition

|               | Formal<br>(1)        | Informal<br>(2)      | Non-emp<br>(3)    | Self-emp<br>(4)   |
|---------------|----------------------|----------------------|-------------------|-------------------|
| Immigration   | 0.271***<br>(0.089)  | −0.284***<br>(0.072) | 0.070<br>(0.075)  | 0.027<br>(0.033)  |
| Baseline Mean | 0.229                | 0.103                | 0.435             | 0.119             |
| Observations  | 3545                 | 3545                 | 3545              | 3545              |
|               | Employer<br>(5)      | Domestic<br>(6)      | Public<br>(7)     | Non-remun<br>(8)  |
| Immigration   | −0.032***<br>(0.012) | −0.029*<br>(0.017)   | −0.015<br>(0.036) | −0.007<br>(0.012) |
| Baseline Mean | 0.019                | 0.046                | 0.038             | 0.012             |
| Observations  | 3,545                | 3,545                | 3,545             | 3,545             |

Notes: IV-drought estimation. Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000.

Table A.3: Labor Market Effects using Hours

|              | Share of hours     |                     |                      | Log hourly wage  |                   |                  |
|--------------|--------------------|---------------------|----------------------|------------------|-------------------|------------------|
|              | Overall<br>(1)     | Formal<br>(2)       | Informal<br>(3)      | Overall<br>(4)   | Formal<br>(5)     | Informal<br>(6)  |
| Immigration  | −0.129*<br>(0.068) | 0.251***<br>(0.088) | −0.380***<br>(0.095) | 0.128<br>(0.272) | −0.418<br>(0.287) | 0.005<br>(0.363) |
| Observations | 3,545              | 3,545               | 3,545                | 3,545            | 3,545             | 3,545            |

Notes: IV-drought estimation. Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000.

looking at non-migrants as well, and by different firm sizes (self-employed, firms with less than 5 employees and firms with at least 6 employees).

Table A.4: Effects on labor force composition at destination

|                                    | Female<br>(1)     | Low Skill<br>(2)  | Young<br>(3)      | Delta Population<br>(4) | Out-Migration<br>(5) |
|------------------------------------|-------------------|-------------------|-------------------|-------------------------|----------------------|
| Panel A: IV-Price<br>Immigration   | −0.085<br>(0.030) | 0.035<br>(0.214)  | 0.261<br>(0.059)  | 0.085<br>(0.514)        | −0.868<br>(0.328)    |
| Observations                       | 3,545             | 3,545             | 3,545             | 3,545                   | 3,545                |
| Panel B: IV-Drought<br>Immigration | −0.019<br>(0.013) | −0.194<br>(0.119) | −0.010<br>(0.020) | 0.478<br>(0.246)        | 0.125<br>(0.111)     |
| Observations                       | 3,545             | 3,545             | 3,545             | 3,545                   | 3,545                |

Notes: IV-drought estimation. Robust standard errors in parenthesis. All regressions have only one regressor: immigration. All regressions are weighted by municipality's population in 2000.

Table A.5: Labor Market Effects by Skill Level

|                                      | Wage employment   |                     |                      | Log monthly wage   |                      |                   |
|--------------------------------------|-------------------|---------------------|----------------------|--------------------|----------------------|-------------------|
|                                      | Overall<br>(1)    | Formal<br>(2)       | Informal<br>(3)      | Overall<br>(4)     | Formal<br>(5)        | Informal<br>(6)   |
| <i>Panel A: High-Skilled Workers</i> |                   |                     |                      |                    |                      |                   |
| Immigration                          | 0.071<br>(0.068)  | 0.330***<br>(0.113) | −0.259***<br>(0.075) | −0.411*<br>(0.217) | −0.797***<br>(0.267) | −0.240<br>(0.336) |
| Observations                         | 3,545             | 3,545               | 3,545                | 3,545              | 3,524                | 3,511             |
| <i>Panel B: Low-Skilled Workers</i>  |                   |                     |                      |                    |                      |                   |
| Immigration                          | −0.013<br>(0.061) | 0.254***<br>(0.076) | −0.267***<br>(0.070) | −0.246<br>(0.379)  | −0.716*<br>(0.420)   | −0.420<br>(0.466) |
| Observations                         | 3,545             | 3,545               | 3,545                | 3,545              | 3,543                | 3,545             |

Notes: IV-drought estimation. Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000.

Table A.6: Labor Market Effects by Migration status

|                              | Wage employment   |                     |                      | Log monthly wage  |                     |                   |
|------------------------------|-------------------|---------------------|----------------------|-------------------|---------------------|-------------------|
|                              | Overall<br>(1)    | Formal<br>(2)       | Informal<br>(3)      | Overall<br>(4)    | Formal<br>(5)       | Informal<br>(6)   |
| <i>Panel A: Non-migrants</i> |                   |                     |                      |                   |                     |                   |
| Immigration                  | −0.025<br>(0.060) | 0.239***<br>(0.082) | −0.263***<br>(0.070) | −0.336<br>(0.322) | −0.845**<br>(0.378) | −0.402<br>(0.382) |
| Observations                 | 3,545             | 3,545               | 3,545                | 3,545             | 3,524               | 3,511             |
| <i>Panel B: Migrants</i>     |                   |                     |                      |                   |                     |                   |
| Immigration                  | 0.034<br>(0.080)  | 0.372***<br>(0.124) | −0.337***<br>(0.080) | 0.452<br>(0.327)  | −0.188<br>(0.291)   | 0.252<br>(0.394)  |
| Observations                 | 3,545             | 3,545               | 3,545                | 3,545             | 3,543               | 3,545             |

Notes: IV-drought estimation. Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000.



Table A.7: Effects on Entrepreneurship by firm size

|                                                          | All                  | Non-migrants         | Migrants          |
|----------------------------------------------------------|----------------------|----------------------|-------------------|
|                                                          | (1)                  | (2)                  | (3)               |
| <i>Panel A: Self-employed</i>                            |                      |                      |                   |
| Immigration                                              | 0.027<br>(0.033)     | 0.011<br>(0.039)     | 0.093*<br>(0.050) |
| Baseline Mean                                            | 0.119                | 0.121                | 0.111             |
| <i>Panel B: Firms with <math>\leq 5</math> employees</i> |                      |                      |                   |
| Immigration                                              | -0.026***<br>(0.009) | -0.034***<br>(0.010) | -0.008<br>(0.012) |
| Baseline Mean                                            | 0.013                | 0.013                | 0.011             |
| <i>Panel C: Firms with <math>\geq 6</math> employees</i> |                      |                      |                   |
| Immigration                                              | -0.006<br>(0.005)    | -0.010<br>(0.006)    | 0.003<br>(0.006)  |
| Baseline Mean                                            | 0.006                | 0.006                | 0.005             |
| Observations                                             | 3,545                | 3,545                | 3,545             |

Notes: IV-drought estimation. Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000.

## B Robustness

Table B.1: Effects of Immigration on Workers - Additional Controls

|                                                               | Wage employment   |                     |                      | Log monthly wage  |                     |                   |
|---------------------------------------------------------------|-------------------|---------------------|----------------------|-------------------|---------------------|-------------------|
|                                                               | Overall<br>(1)    | Formal<br>(2)       | Informal<br>(3)      | Overall<br>(4)    | Formal<br>(5)       | Informal<br>(6)   |
| <i>Panel A: Controlling for Lagged Outcome</i>                |                   |                     |                      |                   |                     |                   |
| Immigration                                                   | -0.029<br>(0.061) | 0.249***<br>(0.077) | -0.274***<br>(0.067) | -0.096<br>(0.240) | -0.751**<br>(0.360) | -0.274<br>(0.348) |
| Observations                                                  | 3,545             | 3,545               | 3,545                | 3,539             | 3,483               | 3,524             |
| <i>Panel B: Controlling for Lagged Population</i>             |                   |                     |                      |                   |                     |                   |
| Immigration                                                   | 0.027<br>(0.062)  | 0.306***<br>(0.103) | -0.279***<br>(0.073) | 0.084<br>(0.189)  | -0.463**<br>(0.210) | 0.115<br>(0.225)  |
| Observations                                                  | 3,545             | 3,545               | 3,545                | 3,545             | 3,545               | 3,545             |
| <i>Panel C: Controlling for Lagged log(Migration)</i>         |                   |                     |                      |                   |                     |                   |
| Immigration                                                   | -0.024<br>(0.066) | 0.264***<br>(0.094) | -0.287***<br>(0.075) | 0.281<br>(0.197)  | -0.386<br>(0.252)   | 0.306<br>(0.276)  |
| Observations                                                  | 3,545             | 3,545               | 3,545                | 3,545             | 3,545               | 3,545             |
| <i>Panel D: Controlling for Lagged log(GDP)</i>               |                   |                     |                      |                   |                     |                   |
| Immigration                                                   | -0.013<br>(0.065) | 0.277***<br>(0.094) | -0.289***<br>(0.073) | 0.190<br>(0.181)  | -0.448*<br>(0.246)  | 0.222<br>(0.249)  |
| Observations                                                  | 3,545             | 3,545               | 3,545                | 3,545             | 3,545               | 3,545             |
| <i>Panel E: Controlling for Lagged Industrial Composition</i> |                   |                     |                      |                   |                     |                   |
| Immigration                                                   | -0.031<br>(0.062) | 0.251***<br>(0.084) | -0.282***<br>(0.074) | -0.091<br>(0.279) | -0.670**<br>(0.330) | -0.176<br>(0.351) |
| Observations                                                  | 3,545             | 3,545               | 3,545                | 3,545             | 3,545               | 3,545             |

Notes: IV-drought estimation. Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000. Panel A has fewer units for wages because data on wages is missing for some municipalities in the census 1991.

Table B.2: Effects of Immigration on Firms - Additional Controls

|                                                               | Nb firms<br>(1)     | Entry<br>(2)        | Exit<br>(3)         | Nb jobs<br>(4)      | Firm wage<br>(5)  |
|---------------------------------------------------------------|---------------------|---------------------|---------------------|---------------------|-------------------|
| <i>Panel A: Controlling for Lagged Outcome</i>                |                     |                     |                     |                     |                   |
| Immigration                                                   | 1.531***<br>(0.319) | 2.561***<br>(0.588) | 2.742***<br>(0.573) | 2.012***<br>(0.607) | -0.764<br>(0.556) |
| Observations                                                  | 3,545               | 3,545               | 3,545               | 3,545               | 3,545             |
| <i>Panel B: Controlling for Lagged Population</i>             |                     |                     |                     |                     |                   |
| Immigration                                                   | 1.631***<br>(0.314) | 2.414***<br>(0.855) | 2.892**<br>(1.153)  | 1.984***<br>(0.642) | -0.487<br>(0.424) |
| Observations                                                  | 3,545               | 3,545               | 3,545               | 3,545               | 3,545             |
| <i>Panel C: Controlling for Lagged log(Migration)</i>         |                     |                     |                     |                     |                   |
| Immigration                                                   | 1.548***<br>(0.337) | 2.802***<br>(1.003) | 3.445***<br>(1.287) | 2.024***<br>(0.662) | -0.118<br>(0.425) |
| Observations                                                  | 3,545               | 3,545               | 3,545               | 3,545               | 3,545             |
| <i>Panel D: Controlling for Lagged log(GDP)</i>               |                     |                     |                     |                     |                   |
| Immigration                                                   | 1.616***<br>(0.327) | 2.759***<br>(0.961) | 3.327***<br>(1.223) | 2.068***<br>(0.639) | -0.244<br>(0.417) |
| Observations                                                  | 3,545               | 3,545               | 3,545               | 3,545               | 3,545             |
| <i>Panel E: Controlling for Lagged Industrial Composition</i> |                     |                     |                     |                     |                   |
| Immigration                                                   | 1.591***<br>(0.306) | 2.619***<br>(0.915) | 2.861**<br>(1.159)  | 1.994***<br>(0.642) | -0.621<br>(0.509) |
| Observations                                                  | 3,545               | 3,545               | 3,545               | 3,545               | 3,545             |

Notes: IV-drought estimation. Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000. Columns 1-3 refer to the log of total number of firms, entrants and exiting firms, respectively.

Table B.3: Effects on Workers – Transformation from destinations to origins following (Borusyak et al., 2022)

|                              | Wage employment     |                      |                       | Log monthly wage  |                     |                   |
|------------------------------|---------------------|----------------------|-----------------------|-------------------|---------------------|-------------------|
|                              | Overall<br>(1)      | Formal<br>(2)        | Informal<br>(3)       | Overall<br>(4)    | Formal<br>(5)       | Informal<br>(6)   |
| <i>Panel A: OLS</i>          |                     |                      |                       |                   |                     |                   |
| Transformed Immigration      | 0.030**<br>(0.014)  | 0.146***<br>(0.014)  | −0.116***<br>(0.009)  | 0.105<br>(0.103)  | −0.049<br>(0.098)   | 0.100<br>(0.125)  |
| Observations                 | 3,545               | 3,545                | 3,545                 | 3,545             | 3,545               | 3,545             |
| <i>Panel B: IV-Drought</i>   |                     |                      |                       |                   |                     |                   |
| Transformed Immigration      | −0.012<br>(0.048)   | 0.273***<br>(0.050)  | −0.285***<br>(0.042)  | −0.126<br>(0.299) | −0.669**<br>(0.271) | −0.198<br>(0.390) |
| F statistic (IV)             | 46.98               | 46.98                | 46.98                 | 46.98             | 46.98               | 46.98             |
| Observations                 | 3,545               | 3,545                | 3,545                 | 3,545             | 3,545               | 3,545             |
| <i>Panel C: Reduced form</i> |                     |                      |                       |                   |                     |                   |
| Drought shock                | −0.0001<br>(0.0004) | 0.002***<br>(0.0005) | −0.003***<br>(0.0003) | −0.001<br>(0.003) | −0.006**<br>(0.002) | −0.002<br>(0.003) |
| Observations                 | 3,545               | 3,545                | 3,545                 | 3,545             | 3,545               | 3,545             |

Notes: IV-drought estimation. Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured at the origin municipality in 2000. The dependent variables and Transformed Immigration are variables at destination (outcomes and the immigration rate) transformed into variables across origins, through a combination of (i) migration patterns between origins and destinations, and (ii) population within destinations at baseline. All regressions are weighted in line with (Borusyak et al., 2022) in the origin municipality.

Table B.4: Effects on Firms – Transformation from destinations to origins following (Borusyak et al., 2022)

|                              | Nb firms<br>(1)     | Entry<br>(2)        | Exit<br>(3)         | Nb jobs<br>(4)      | Firm wage<br>(5)  |
|------------------------------|---------------------|---------------------|---------------------|---------------------|-------------------|
| <i>Panel A: OLS</i>          |                     |                     |                     |                     |                   |
| Transformed Immigration      | 1.583***<br>(0.067) | 1.429***<br>(0.238) | 1.962***<br>(0.378) | 1.213***<br>(0.129) | 0.295<br>(0.186)  |
| Observations                 | 3,545               | 3,545               | 3,545               | 3,545               | 3,545             |
| <i>Panel B: IV-Drought</i>   |                     |                     |                     |                     |                   |
| Transformed Immigration      | 1.621***<br>(0.187) | 2.550***<br>(0.841) | 2.765**<br>(1.263)  | 2.005***<br>(0.337) | −0.770<br>(0.509) |
| F statistic (IV)             | 46.98               | 46.98               | 46.98               | 46.98               | 46.98             |
| Observations                 | 3,545               | 3,545               | 3,545               | 3,545               | 3,545             |
| <i>Panel C: Reduced form</i> |                     |                     |                     |                     |                   |
| Drought shock                | 0.014***<br>(0.003) | 0.023***<br>(0.008) | 0.025**<br>(0.012)  | 0.018***<br>(0.004) | −0.007<br>(0.005) |
| Observations                 | 3,545               | 3,545               | 3,545               | 3,545               | 3,545             |

Notes: IV-drought estimation. Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured at the origin municipality in 2000. The dependent variables and Transformed Immigration are variables at destination (outcomes and the immigration rate) transformed into variables across origins, through a combination of (i) migration patterns between origins and destinations, and (ii) population within destinations at baseline. All regressions are weighted in line with (Borusyak et al., 2022) in the origin municipality. Columns 1-3 refer to the log of total number of firms, entrants and exiting firms, respectively.

Table B.5: Effects of Immigration on Workers - Alternative Channels

|                                                                                      | Wage employment   |                     |                      | Log monthly wage   |                      |                     |
|--------------------------------------------------------------------------------------|-------------------|---------------------|----------------------|--------------------|----------------------|---------------------|
|                                                                                      | Overall<br>(1)    | Formal<br>(2)       | Informal<br>(3)      | Overall<br>(4)     | Formal<br>(5)        | Informal<br>(6)     |
| <i>Panel A: Controlling for local and neighboring municipalities' drought shocks</i> |                   |                     |                      |                    |                      |                     |
| Immigration                                                                          | -0.017<br>(0.055) | 0.215***<br>(0.074) | -0.231***<br>(0.053) | -0.470*<br>(0.242) | -0.941***<br>(0.286) | -0.675**<br>(0.299) |
| Observations                                                                         | 3,545             | 3,545               | 3,545                | 3,545              | 3,545                | 3,545               |
| <i>Panel B: Controlling for capital reallocation channel</i>                         |                   |                     |                      |                    |                      |                     |
| Immigration                                                                          | -0.036<br>(0.059) | 0.244***<br>(0.082) | -0.279***<br>(0.071) | -0.163<br>(0.270)  | -0.656**<br>(0.321)  | -0.259<br>(0.331)   |
| Observations                                                                         | 2,627             | 2,627               | 2,627                | 2,627              | 2,627                | 2,627               |

Notes: IV-drought estimation. Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000. Panel B has 2627 units because data on banks is missing for 918 municipalities.

Table B.6: Effects of Immigration on Firms - Alternative Channels

|                                                                                      | Nb firms<br>(1)     | Entry<br>(2)        | Exit<br>(3)        | Nb jobs<br>(4)      | Firm wage<br>(5)    |
|--------------------------------------------------------------------------------------|---------------------|---------------------|--------------------|---------------------|---------------------|
| <i>Panel A: Controlling for local and neighboring municipalities' drought shocks</i> |                     |                     |                    |                     |                     |
| Immigration                                                                          | 1.941***<br>(0.295) | 1.701**<br>(0.676)  | 1.251<br>(0.948)   | 2.016***<br>(0.571) | −1.020**<br>(0.497) |
| Observations                                                                         | 3,545               | 3,545               | 3,545              | 3,545               | 3,545               |
| <i>Panel B: Controlling for capital reallocation channel</i>                         |                     |                     |                    |                     |                     |
| Immigration                                                                          | 1.547***<br>(0.298) | 2.653***<br>(0.893) | 2.771**<br>(1.129) | 1.984***<br>(0.611) | −0.704<br>(0.527)   |
| Observations                                                                         | 2,627               | 2,627               | 2,627              | 2,627               | 2,627               |
| <i>Panel C: Excluding firms that produce agricultural goods</i>                      |                     |                     |                    |                     |                     |
| Immigration                                                                          | 2.018***<br>(0.356) | 2.991***<br>(0.935) | 2.589**<br>(1.136) | 2.080***<br>(0.640) | −0.946<br>(0.593)   |
| Observations                                                                         | 3,545               | 3,545               | 3,545              | 3,545               | 3,545               |

Notes: IV-drought estimation. Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000. Panel B has 2627 units because data on banks is missing for 918 municipalities. Columns 1-3 refer to the log of total number of firms, entrants and exiting firms, respectively.

## B.1 The Price Push Shock

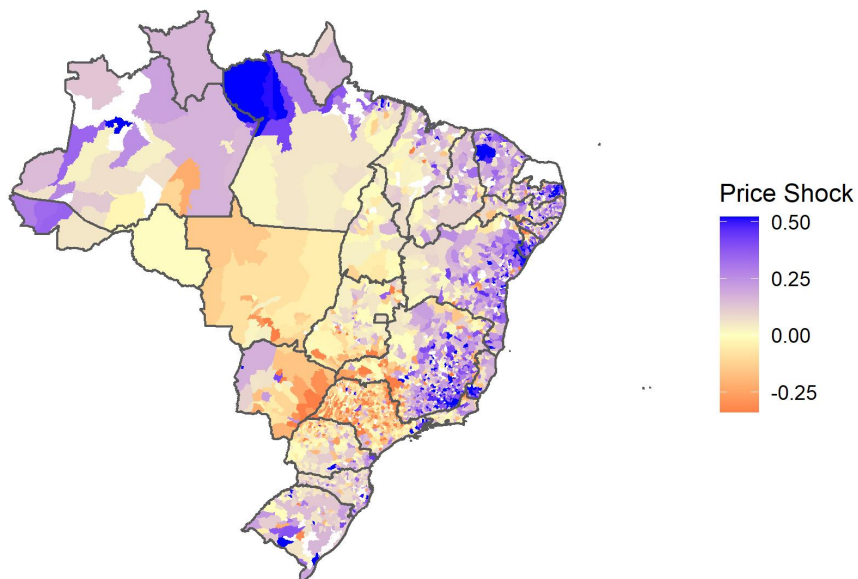
### B.1.1 Construction of price push shock

We collect monthly information on international prices for 12 crops (including bananas, cocoa, coffee, cotton, maize, orange, rice, soybeans, sugar, tobacco, wheat and wood) between 1970 and 2011. For each crop  $c$  and month  $m$ , we compute a price shock  $\varepsilon_{cm}$  as the residual of an AR(1) process. We then aggregate these crop- and month-level shocks into a (origin) municipality level shock  $s_o^{price}$  using the share of each crop  $\pi_{oc}$  in the value of agricultural production in the municipality of origin  $o$ . We use the 1980 Agricultural Census to compute these shares for each municipality and crop. Formally, the price shock is given by:

$$s_o^{prices} = \sum_m \sum_c (\pi_{oc} \times \varepsilon_{cm}) \quad (17)$$

Figure B.1 shows the geographic distribution of the price shock.

Figure B.1: Push Shocks: Crop Prices



Notes: The Price shocks are constructed according to expression 17.



### B.1.2 Main results using the price shock

Analogously to our main specification, we combine cross-sectional variation across destination municipalities in their pre-existing migration networks with different origins (the “share”), and time variation in exogenous push shocks that affect migration incentives at origin (the “shift”). Formally, the instrument writes:

$$Z_d^{prices} = \sum_o \lambda_{o,d} s_o^{prices} \quad (18)$$

where  $Z_d^{prices}$  denotes the instrument for immigration into the municipality of destination  $d$ , based on the price shock  $s_o^{prices}$  to agricultural productivity in the municipality of origin  $o$ . Again,  $\lambda_{o,d}$  denotes the share of migrants from origin  $o$  among migrants who had come at destination  $d$  between 1995 and 2000. We then use  $Z_d^{prices}$  as an instrument for  $Mig_d$  in a 2SLS estimator.

Table B.7: Effects of Immigration on Workers – Price Shocks

|                   | Wage employment  |                  |                   | Log monthly wage  |                   |                   |
|-------------------|------------------|------------------|-------------------|-------------------|-------------------|-------------------|
|                   | Overall<br>(1)   | Formal<br>(2)    | Informal<br>(3)   | Overall<br>(4)    | Formal<br>(5)     | Informal<br>(6)   |
| Panel A: OLS      |                  |                  |                   |                   |                   |                   |
| Immigration       | 0.037<br>(0.019) | 0.105<br>(0.023) | −0.068<br>(0.014) | 0.062<br>(0.076)  | 0.032<br>(0.068)  | 0.035<br>(0.092)  |
| Panel B: IV-Price |                  |                  |                   |                   |                   |                   |
| Immigration       | 0.100<br>(0.103) | 0.394<br>(0.149) | −0.294<br>(0.100) | −1.599<br>(0.580) | −2.168<br>(0.680) | −1.909<br>(0.743) |
| F Statistic (IV)  | 16.5             | 16.5             | 16.5              | 16.5              | 16.5              | 16.5              |
| Baseline Mean     | 0.332            | 0.229            | 0.103             | –                 | –                 | –                 |
| Observations      | 3,545            | 3,545            | 3,545             | 3,545             | 3,545             | 3,545             |

Notes: Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality’s population in 2000.

Table B.8: Effects of Immigration on Firms – Price Shocks

|                     | Nb firms<br>(1)  | Entry<br>(2)     | Exit<br>(3)      | Nb jobs<br>(4)   | Firm wage<br>(5)  |
|---------------------|------------------|------------------|------------------|------------------|-------------------|
| Panel A: OLS        |                  |                  |                  |                  |                   |
| Immigration         | 1.342<br>(0.108) | 1.167<br>(0.203) | 1.588<br>(0.274) | 1.071<br>(0.270) | 0.372<br>(0.101)  |
| Panel B: IV - Price |                  |                  |                  |                  |                   |
| Immigration         | 2.354<br>(0.622) | 7.526<br>(2.085) | 5.583<br>(2.045) | 2.176<br>(0.855) | −3.436<br>(1.169) |
| F Statistic (IV)    | 16.5             | 16.5             | 16.5             | 16.5             | 16.5              |
| Observations        | 3,545            | 3,545            | 3,545            | 3,545            | 3,545             |

Notes: Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000. Columns 1-3 refer to the log of total number of firms, entrants and exiting firms, respectively.

## C Mechanisms

### C.1 Wage rigidity

Table C.1: Wage Rigidity and Labor Market Effects

|                           | Formal Employment   |                      |                     | Informal Employment  |                      |                      |
|---------------------------|---------------------|----------------------|---------------------|----------------------|----------------------|----------------------|
| <i>Panel A</i>            | (1)                 | (2)                  | (3)                 | (1)                  | (2)                  | (3)                  |
| Immigration               | 0.271***<br>(0.089) | 0.270***<br>(0.091)  | 0.336***<br>(0.106) | −0.284***<br>(0.072) | −0.284***<br>(0.073) | −0.294***<br>(0.081) |
| × Binding MW              |                     | −0.385***<br>(0.071) |                     |                      | 0.086**<br>(0.044)   |                      |
| × High DNWR               |                     |                      | −0.097*<br>(0.050)  |                      |                      | 0.015<br>(0.036)     |
| Pr(IMR + IMR × [Het] = 0) |                     | 0.0271               | <0.0001             |                      | <0.0001              | <0.0001              |
| Observations              | 3,545               | 3,545                | 3,545               | 3,545                | 3,545                | 3,545                |

|                           | Nonemployment    |                     |                    | Formal Wages        |                     |                      |
|---------------------------|------------------|---------------------|--------------------|---------------------|---------------------|----------------------|
| <i>Panel B</i>            | (1)              | (2)                 | (3)                | (1)                 | (2)                 | (3)                  |
| Immigration               | 0.070<br>(0.075) | 0.072<br>(0.069)    | 0.019<br>(0.085)   | −0.671**<br>(0.336) | −0.671**<br>(0.338) | −1.129***<br>(0.438) |
| × Binding MW              |                  | 0.564***<br>(0.074) |                    |                     | 0.151<br>(0.320)    |                      |
| × High DNWR               |                  |                     | 0.076**<br>(0.038) |                     |                     | 0.679***<br>(0.168)  |
| Pr(IMR + IMR × [Het] = 0) |                  | <0.0001             | 0.0031             |                     | 0.0155              | 0.0003               |
| Observations              | 3,545            | 3,545               | 3,545              | 3,545               | 3,545               | 3,545                |

Notes: IV-drought estimation. *Binding MW* is a dummy that equals one if the municipality's Kaitz index (the ratio between the national minimum wage and the municipality's median wage) is above the national median. *High DNWR* is a dummy that equals one if the municipality's share of workers with negative year-on-year hourly wage changes among workers with non-zero wage changes is below the median share. Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000.

### C.2 Short-Run Effects

For the short run (year-on-year) specification, we explore the higher frequency of the shock and aggregate up to year  $t$ , so that the instrument is now given by:

$$s_{o,t}^{drought} = \sum_{m=t_0}^t \sum_c (\pi_{oc} \times g_{ocm} \times D_{om}) \quad (19)$$

We construct the instrument in a similar way to the one described in Section 3.1 (expression 3), but we fully explore the annual variation in the drought shocks:

$$Z_{d,t} = \sum_o \lambda_{o,d} s_{o,t}^{drought}$$

where  $s_{o,t}^{drought}$  is given by expression 19. We instrument  $Mig_{dt}$  with  $Z_{d,t-1}$ .

Table C.2 presents the results for the short-run effects of immigration on formal and informal wage employment, unemployment, and formal wages.

Table C.2: Short-Run Effects

|                  | Formal Employment   | Informal Employment | Nonemployment    | Formal Wages      |
|------------------|---------------------|---------------------|------------------|-------------------|
|                  | (1)                 | (2)                 | (3)              | (4)               |
| Immigration      | -1.198**<br>(0.584) | 0.027<br>(0.303)    | 0.209<br>(0.520) | -1.202<br>(2.324) |
| F statistic (IV) | 21.53               | 21.53               | 21.53            | 21.53             |
| Observations     | 6,407               | 6,407               | 6,407            | 6,407             |

Notes: IV-drought estimation. Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000.

### C.3 Long Differences (Decadal Changes) using Droughts as Push Shocks and Municipalities Available in PNAD

In this subsection, we re-estimate our main results discussed in Section 3, but restricting the sample to the 700 municipalities identifiable in the National Household Survey (PNAD).

Table C.3: Long-Run Effects on Workers – Drought Shocks with PNAD Municipalities

|                     | Wage employment    |                     |                      | Log monthly wage  |                    |                   |
|---------------------|--------------------|---------------------|----------------------|-------------------|--------------------|-------------------|
|                     | Overall            | Formal              | Informal             | Overall           | Formal             | Informal          |
|                     | (1)                | (2)                 | (3)                  | (4)               | (5)                | (6)               |
| Panel A: OLS        |                    |                     |                      |                   |                    |                   |
| Immigration         | 0.004<br>(0.022)   | 0.067***<br>(0.023) | −0.062***<br>(0.018) | 0.059<br>(0.099)  | 0.024<br>(0.086)   | 0.044<br>(0.124)  |
| Panel B: IV-Drought |                    |                     |                      |                   |                    |                   |
| Immigration         | −0.114*<br>(0.060) | 0.104*<br>(0.056)   | −0.218***<br>(0.061) | −0.349<br>(0.306) | −0.629*<br>(0.355) | −0.385<br>(0.371) |
| F Statistic (IV)    | 19.3               | 19.3                | 19.3                 | 19.3              | 19.3               | 19.3              |
| Baseline Mean       | 0.338              | 0.243               | 0.095                | -                 | -                  | -                 |
| Observations        | 700                | 700                 | 700                  | 700               | 700                | 700               |

Notes: IV-drought estimation. Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000.

Table C.4: Long-Run Effects on Firms – Drought Shocks with PNAD Municipalities

|                     | Nb firms            | Entry               | Exit                | Nb jobs             | Firm wage           |
|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                     | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 |
| Panel A: OLS        |                     |                     |                     |                     |                     |
| Immigration         | 1.215***<br>(0.119) | 1.296***<br>(0.283) | 1.870***<br>(0.365) | 0.922***<br>(0.319) | 0.498***<br>(0.134) |
| Panel B: IV-Drought |                     |                     |                     |                     |                     |
| Immigration         | 1.225***<br>(0.279) | 2.503***<br>(0.843) | 2.678**<br>(1.125)  | 1.135**<br>(0.446)  | −0.276<br>(0.535)   |
| F Statistic (IV)    | 19.3                | 19.3                | 19.3                | 19.3                | 19.3                |
| Observations        | 700                 | 700                 | 700                 | 700                 | 700                 |

Notes: IV-drought estimation. Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000.

## D Model Appendix

This section contains the propositions and corollaries discussed in Section 4 and their respective proofs. For notational simplicity, we discuss the case without heterogeneous growth profiles, that is, without a firm-specific intercept in the productivity process.

**Proposition 1 (*existence and uniqueness of the stationary equilibrium*):**

*Under the assumptions that (i) the general law of motion of the productivity process in both sectors, denoted here by  $G(\theta'|\theta)$ , is continuous in both arguments and strictly decreasing in  $\theta$ ; and (ii) the profit functions are such that  $q_s$  and  $\ell_s$  are continuous, single-valued and strictly increasing in  $\theta$  (as in 6 and 7). Then, there exists a unique stationary competitive equilibrium where both the formal and informal sector exist, and both sectors have positive entry and exit.*

Proof: The proof of this proposition is organized in several steps.

*Step 1: Properties of the profit and value functions*

The assumption about the production and cost functions in both sectors implies that the profit functions are continuous and that  $q_j^*$  and  $l_j^*$  are continuous, single valued, and strictly increasing in  $\theta$ . Write the corresponding equilibrium prices as

$$w^*(\mu) = W\left(L^*(\mu, w(\mu))\right) \quad (20)$$

It follows directly from Hopenhayn (1992), Lemma 3, that the function  $w^*(\mu)$  is well defined and continuous. Using 20, one can re-write the value functions in each sector as follows:

$$V_f(\theta, \mu) = \tilde{\pi}_f(\theta, \mu) + (1 - \delta_f)\beta \max\{0, E[V_f(\theta', \mu) | \theta]\} \quad (21)$$

$$V_i(\theta, \mu) = \tilde{\pi}_i(\theta, \mu) + \beta \max\{0, (1 - \delta_i)E[V_i(\theta', \mu) | \theta], (1 - \delta_f)E[V_f(\theta', \mu) | \theta] - \tilde{c}^e\} \quad (22)$$

where  $\tilde{\pi}_s(\theta, \mu) \equiv \pi_s(\theta, w^*(\mu))$ .

LEMMA 1: *The functions  $\tilde{\pi}_s$  are continuous in both arguments, strictly increasing in  $\theta$  and decreasing in  $\mu$ .*

Proof: The proof follows directly from the proof in Hopenhayn (1992). Continuity follows from the continuity of the profit functions in  $\theta$  and  $w$ , and from continuity of

$w^*(\mu)$ . Since  $p^*(\mu) > 0, \forall \mu$ , it also follows from the properties of the profit function that  $\tilde{\pi}_s$  is increasing in  $\theta$ .

It remains to show that  $\tilde{\pi}_j$  is decreasing in  $\mu$ . Let  $\mu_2 > \mu_1$  with the corresponding equilibrium prices  $w_1^*$  and  $w_2^*$ . Suppose by way of contradiction that  $w_2^* < w_1^*$ . This implies that  $q_s(\theta, w_2^*) > q_s(\theta, w_1^*) \forall \theta$  and therefore  $Q_s^*(\mu_2) = \int q_s(\theta, w_2^*) d\mu_2^j(\theta) > \int q_s(\theta, w_1^*) d\mu_1^j(\theta) = Q_s^*(\mu_1)$  (the last inequality follows because  $q_j$  is strictly increasing in  $\theta$ ),  $s = i, f$ . But if  $Q^*(\mu_2) > Q^*(\mu_1)$ , then  $L^*(\mu_2) > L^*(\mu_1)$  and therefore  $w_2^* > w_1^*$  (as labor supply is fixed), which contradicts the initial assumption that  $w_2^* < w_1^*$ . Hence, if  $\mu_2 > \mu_1$  then  $w_2^* > w_1^*$ . Hence, one can easily verify that  $\tilde{\pi}_s(\theta, \mu_2) \equiv \tilde{\pi}_s(\theta, w_2^*) < \tilde{\pi}_s(\theta, w_1^*) \equiv \tilde{\pi}_s(\theta, \mu_1)$ , which establishes the desired result.  $\square$

Given these properties of the profit functions, we can go ahead and establish the properties of the value functions:

**LEMMA 2:** *The functions,  $V_s$ , that solve [21](#) and [22](#) are unique and have the following properties: (i) they are continuous functions; (ii) strictly increasing in  $\theta$  and decreasing in  $\mu$ ; and (iii) the option value in both sectors (the integral terms in [21](#) and [22](#)) is strictly increasing in  $\theta$ .*

*Proof:* Given the assumptions about the productivity process and the properties of  $\tilde{\pi}_j$  established in Lemma 1, the properties (i) and (ii) follow directly from standard dynamic programming arguments. The same can be said about (iii), except that it also relies on (ii). Note that the presence of  $V_f$  in the option value of the informal firm does not alter the argument, as the value functions in both sectors share the same properties.  $\square$

*Step 3: Uniqueness of the cut-offs for entry, exit and informal to formal transitions.*

Due to the properties of the value functions established in Lemma 2, it follows that  $\underline{\theta}_s, \underline{\nu}_s$ , and  $\bar{\theta}_i$ , are uniquely determined by the following expressions:

$$\int V_s(\theta', \mu) dF(\theta' | \underline{\theta}_s) = 0, \quad s = i, f \quad (23)$$

$$\int V_i(\theta, z) dF(\theta | \underline{\nu}_i) = c_i^e \quad (24)$$

$$\int V_f(\theta, z) dF(\theta | \underline{\nu}_f) = V_i^e(\underline{\nu}_f, w) - (c_i^e - c_f^e) \quad (25)$$

$$\int [V_f(\theta', \mu) - V_i(\theta', \mu)] dF(\theta' | \bar{\theta}_i) = \tilde{c}^e \quad (26)$$

Using assumption (A.2.ii) and the functional forms assumed for the production and cost functions in each sector, one can verify that  $\underline{\theta}_f > \underline{\theta}_i$ . This will be the case even

if at  $\theta = \underline{\theta}_f$  the formal firm hires all of its employees informally. Using a similar reasoning,  $c_f^e > c_i^e$  implies that  $\underline{\nu}_f > \underline{\nu}_i$ .

Finally, condition 26 requires a more careful explanation. There is a productivity threshold  $\bar{\theta}$  above which  $\pi_f(\theta) > \pi_i(\theta)$ , for any  $\theta > \bar{\theta}$ . Hence, from the results proved in Lemmas 1 and 2 it follows that, if there is informal to formal transition, then the threshold  $\bar{\theta}_i$  defined by 26 is unique.

*Step 4: The industry measures*

Following Hopenhayn (1992), we define the operator  $\hat{P}_k$  as follows:

$$\hat{P}_k \equiv \hat{P}_k(\theta, B) = \begin{cases} \int_B dF(x|\theta) & \text{if } \theta \in I_k \\ 0 & \text{otherwise} \end{cases}$$

for all Borel sets  $B \subset \Theta$ ; where  $I_1 = [\underline{\theta}_i, \bar{\theta}_i)$ ,  $I_2 = [\bar{\theta}_i, \infty)$ , and  $I_3 = [\underline{\theta}_f, \infty)$ . This is a bounded ( $\|\hat{P}_k\| \leq 1$ ), linear operator on the space of positive bounded measures:  $\hat{P}_k \mu(B) = \int \hat{P}_k(\theta, B) d\mu(\theta)$ , for all Borel sets  $B \subset \Theta$  (see Hopenhayn, 1992). Using this notation, one can rewrite the law of motion of both sectors' measure as follows:

$$\mu'_i = \hat{P}_1 \mu_i + \Lambda'_i \tilde{G}'_i \quad (27)$$

$$\mu'_f = \hat{P}_3 \mu_f + \Lambda'_f \tilde{G}'_f + \hat{P}_2 \mu_i \quad (28)$$

The properties of the productivity process and the definition of the sets  $I_k$  imply that  $\|\hat{P}_k\| < 1$ . Hence, the argument presented in Hopenhayn (1992, Lemma 4) follows directly and the operator  $(I - \hat{P}_k)$  has an inverse. Hence, the invariant measures  $\mu_i$  and  $\mu_f$  are well-defined and can be written as

$$\mu_i = (I - \hat{P}_1)^{-1} \Lambda_i \tilde{G}_i \quad (29)$$

$$\mu_f = (I - \hat{P}_3)^{-1} (\Lambda_f \tilde{G}_f + \hat{P}_2 \mu_i) \quad (30)$$

Write the above measures as  $\mu_i = m_i(\underline{\theta}_i, \bar{\theta}_i, \Lambda_i)$  and  $\mu_f = m_f(\underline{\theta}_f, \bar{\theta}_i, \Lambda_f)$ , where we use the fact that the operators  $\hat{P}_k$  are functions of the cut-offs that define the sets  $I_k$ . It is then useful to establish the following result:

**LEMMA 3:** *The functions  $m_i(\underline{\theta}_i, \bar{\theta}_i, \Lambda_i)$  and  $m_f(\underline{\theta}_f, \bar{\theta}_i, \Lambda_f)$  are continuous in all of their arguments. Moreover,  $m_i(\underline{\theta}_i, \bar{\theta}_i, \Lambda_i)$  is strictly increasing in  $\Lambda_i$  and  $m_f(\underline{\theta}_f, \bar{\theta}_i, \Lambda_f)$  is strictly increasing in  $\Lambda_f$ .*

*Proof:* From expressions 29 and 30, it is clear that the functions  $m_s(\cdot)$  are continuous and strictly increasing in  $\Lambda_s$ . The continuity in the cut-offs is a consequence of the



fact that the operator  $\hat{P}_k$  is continuous in the cut-offs that define the corresponding set  $I_k$  (see [Hopenhayn, 1992](#), Lemma 5 for a proof).  $\square$

#### *Step 5: Existence and uniqueness*

The starting point is to use the following results established in [Hopenhayn \(1992\)](#): (i) there exists a stationary equilibrium with invariant measure  $\mu$  associated to an unique aggregate input-output pair,  $(L, Q)$ , and unique prices (Theorem 2); (ii) if the entry cost is low enough, a stationary equilibrium with positive entry exists (Theorem 3); (iii) if the profit function is multiplicatively separable between productivity and prices,  $\pi(\theta, p, w) = h(\theta)g(p, w)$ , then if there exists an equilibrium with entry and exit, it is unique (Theorem 4).<sup>25</sup> It remains to show that there exists an unique stationary equilibrium with an unique formal-informal partition and entry into both sectors.

First, write  $\mu = m(\underline{\theta}_i, \bar{\theta}_i, \underline{\theta}_f, \Lambda_f, \Lambda_i) = m_i(\underline{\theta}_i, \bar{\theta}_i, \Lambda_i) + m_f(\underline{\theta}_f, \bar{\theta}_i, \Lambda_f)$ . Fix  $\mu$ ; from Step 3 (equations 23-26) we know that there are unique cut-off points for entry, exit and transition between sectors. In particular, because  $c_f^e > c_i^e$  the following holds:  $\underline{\nu}_f > \underline{\nu}_i$ . Thus, for any  $\nu \in [\underline{\nu}_i, \underline{\nu}_f)$ , entry occurs into the informal sector and for any  $\nu \geq \underline{\nu}_f$ , entry occurs into the formal sector. Hence, the unique thresholds  $(\underline{\nu}_i, \underline{\nu}_f)$  pin down the mass of entrants into the informal sector,  $\Lambda_i$ . Additionally, fixing  $\mu$  uniquely determines thresholds  $(\underline{\theta}_i, \bar{\theta}_i)$ , which therefore pins down a unique informal sector size,  $\mu_i = m_i(\underline{\theta}_i, \bar{\theta}_i, \Lambda_i)$ .

Finally, the industry size  $\mu$  also uniquely determines the formal sector's threshold,  $\underline{\theta}_f$ . But  $\mu = \mu_i + \mu_f$  and  $\mu_f = m_f(\underline{\theta}_f, \bar{\theta}_i, \Lambda_f)$  is strictly increasing in  $\Lambda_f$ . Thus, once the informal sector size is determined and the thresholds  $(\underline{\theta}_f, \bar{\theta}_i)$  are fixed, there is an unique value of  $\Lambda_f$  that satisfies the identity  $\mu = \mu_i + \mu_f$ .

Thus, there is an unique stationary equilibrium with invariant  $\mu_s$ ,  $(\underline{\nu}_s, \underline{\theta}_s, \bar{\theta}_i)$ ,  $s = i, f$ , aggregate prices and quantities,  $(Q, L, w)$ , and entry levels in both sectors,  $(\Lambda_f, \Lambda_i)$ .

$\square$

### **The life cycle of firms: size, productivity and informality**

Let the productivity distribution among firms of age  $n$  in sector  $s$  be denoted by the probability measure  $\lambda_s^n$ . This measure is defined over the set of active firms with a given age,  $A_s^n$ . The evolution of the productivity distribution in the informal and

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<sup>25</sup>The functional form assumed for the production function belongs to this class and thus the theorem applies directly.

formal sectors within a given cohort is given by:

$$\begin{aligned}\lambda_i^{n+1}(\theta') &= \int_{X_1} G(\theta'|\theta) d\lambda_i^n(\theta) \\ \lambda_f^{n+1}(\theta') &= \int_{X_2} G(\theta'|\theta) d\lambda_f^n(\theta) + \int_{X_3} G(\theta'|\theta) d\lambda_i^n(\theta)\end{aligned}$$

for all  $\theta' \in \Theta$ ;  $X_1 \equiv A_i^n \cap [\underline{\theta}_i, \bar{\theta}_i)$ ,  $X_2 \equiv A_f^n \cap [\underline{\theta}_f, \infty)$ , and ,  $X_3 \equiv A_i^n \cap [\bar{\theta}_i, \infty)$ .

Now consider the ordering of measures given by the the first order stochastic dominance criterion, so that  $\lambda_s^{n+1} \succeq \lambda_s^n$  means that the productivity distribution at age  $n+1$  first order stochastically dominates the one at age  $n$ . Then the following result holds:

**Proposition 2:** *Assume that the assumptions of Proposition 1 hold and that the pre-entry signal parameter has a continuous distribution,  $H(\nu)$ . Then the productivity distribution among active firms within a sector and a cohort is increasing in the cohort's age. That is,  $\lambda_s^{n+1} \succeq \lambda_s^n$  for all  $n$  and  $s = i, f$ .*

Proof:

Define the following operators:

$$T_k(B) = \begin{cases} \int_B dF(x|\theta), & \text{if } \theta \in X_k \\ 0 & \text{otherwise} \end{cases}$$

for  $k = 1, 2, 3$  and all Borel sets  $B \subseteq \Theta$ . As before,  $X_1 \equiv A_i^n \cap [\underline{\theta}_i, \bar{\theta}_i)$ ,  $X_2 \equiv A_f^n \cap [\underline{\theta}_f, \infty)$ , and ,  $X_3 \equiv A_i^n \cap [\bar{\theta}_i, \infty)$ . One can then rewrite the expressions for the  $\lambda_s^n$  as follows:

$$\begin{aligned}\lambda_i^{n+1} &= T_1 \lambda_i^n \\ \lambda_f^{n+1} &= T_2 \lambda_f^n + T_3 \lambda_i^n\end{aligned}$$

The productivity distribution of newborn firms, however, is simply given by  $\lambda_i^1([\theta_l, \infty)) = \int_{\nu > \underline{\nu}_i} F(\theta|\nu) dG(\nu)$  and  $\lambda_f^1([\theta_l, \infty)) = \int_{\nu > \underline{\nu}_f} F(\theta|\nu) dG(\nu)$ .

From the second period on, the productivity distribution is obtained by first applying the truncation implied by the conditions  $\theta \in X_k$  in the operators  $T_k$ . This always implies a truncation on the lower tail of the productivity distribution, which is increasing in the FOSD ordering. Finally, assumption (A.2) implies that after the truncation, the operator  $T_k$  is also monotone and hence the operator  $T_k$  as defined is increasing in the FOSD criterion. Thus, the following holds almost by definition:  $\lambda_i^2 \equiv T_1 \lambda_i^1 \succeq \lambda_i^1$ . By induction,  $\lambda_s^{n+1} \succeq \lambda_s^n$  for any  $n$ .

The analysis of  $\lambda_f^n$  is not so straightforward because of the presence of the term

$T_3\lambda_i^n$ , so a more careful argument is needed. First, note that because of the argument just made for  $\lambda_i^n$ , in the absence of the term  $T_3\lambda_i^n$  one would observe  $\lambda_f^{n+1} \succeq \lambda_f^n$  for any  $n$ . Second, the conditioning embedded in operator  $T_3$  is stronger than the one in  $T_2$ , as  $\bar{\theta}_i > \underline{\theta}_f$ . Thus,  $\lambda_f^2 = T_2\lambda_f^1 + T_3\lambda_i^1 \succeq \lambda_f^1$ . But given the result that  $\lambda_i^n$  is increasing in  $n$ , and that the  $T_k$  operators are increasing, then  $\lambda_f^n$  will be also increasing in  $n$ .  $\square$

**Corollary 1:** *As the productivity distribution among active firms within a sector and a cohort is increasing in the cohort's age, so is the integral of any function that is increasing in  $\theta$ . In particular, this is true for the survival rate, average size and average revenue.*

Proof: Given the result in Proposition 2, this corollary follows mechanically and no proof is provided.

**Corollary 2 (intensive margin of informality):** *As a consequence of Proposition 2 and Corollary 1, the average informality rate within formal firms in a given cohort is decreasing in cohort's age.*

Proof: There is an unique threshold  $\tilde{\ell}$  above which the formal firm only hires formal workers (on the margin). For a formal firm that has its optimal level of labor below the threshold,  $\ell_f^*(\theta) < \tilde{\ell}$ ,  $s_i(\theta) = 1$ . In this case, the within informality rate will stay constant at one for some period while the firm is growing, but as soon as  $\ell_f^*(\theta) = \tilde{\ell}$  the informal share will start declining monotonically with firm's size. Similarly, for any initial value of  $s_i < 1$ , as the firm grows the  $s_i$  will decline monotonically. Hence, the  $s_i$  is a constant function of  $\theta$  if  $\ell(\theta) < \tilde{\ell}(\theta)$  and it is strictly decreasing in  $\theta$  over the range where  $\ell(\theta) \geq \tilde{\ell}(\theta)$ . Combined with the result in Proposition 2, this implies that the average within firm informality for a given cohort will be monotonically decreasing with cohort's age almost everywhere in the relevant range  $[\underline{\theta}_f, \infty]$ . As long as there is always at least one firm within the cohort that has a  $s_i(\theta) < 1$ , the average within firm informality will be strictly decreasing in the cohort's age.  $\square$

It is straightforward to see that the share of informal workers is decreasing in firm's size (and therefore productivity) in the within-period (static) problem of the firm. Thus, by Proposition 1 and Corollary 1 one gets Corollary 2. Finally, let the informality rate among firms of a given cohort of age  $n$  be expressed as  $B^n = \frac{\mu(A_i^n)}{\mu(A_i^n) + \mu(A_f^n)}$ , where  $\mu(A_s^n)$  denotes the measure of active firms in sector  $s$  with age  $n$ . With this notation in hand, the last result of this section can be stated as follows:

**Corollary 3 (extensive margin of informality)** *Under the conditions of Proposition 2, the informality rate  $B^n$  within a given cohort is weakly decreasing in cohort's age. That is,  $B^{n+1} \leq B^n$  for all  $n$ .*

Proof: Proposition 2 established that the distribution of productivity is increasing with firms' age in both sectors. At the same time, the informal to formal transition threshold,  $\bar{\theta}_i$  remains constant. Hence, as the cohort gets older a smaller number of

firms (among the survivors) will remain informal as the most productive ones keep making the transition into the formal sector. This implies that  $\mu_i^{n+1} \leq \mu_i^n$ . The same is not true for  $\mu_f^n$ , as there is no upper limit for formal stayers. This means that formal firms do not face the additional exit margin that informal firms do, as there is no upper limit for their growth. Since the productivity shock in both sectors is the same, even if the set of active firms in the formal sector reduces in size as the cohort ages, it does so at a lower rate than informal firms in the same cohort. Hence, the ratio  $\gamma_b^n = \frac{\mu_i^n}{\mu_i^n + \mu_f^n}$  is weakly decreasing in the cohort's age.  $\square$

## E Model Calibration Appendix

### E.1 Estimating formal sector's persistence parameter

One can use the first order condition of formal firms that hire at least one formal worker and the law of motion of formal firms' log-productivity to show that the establishment-level employment process can be represented as a simple AR(1) process with the same persistence parameter,  $\rho_f$ . This procedure assumes that there are no adjustment costs, such as hiring costs, which could lead to an overestimation of  $\rho_f$ . However, [Dix-Carneiro et al. \(2024\)](#) estimate this persistence parameter in a richer model with adjustment costs and find slightly higher (and not lower) values.

Let the log of firm  $j$ 's employment at time  $t$  be denoted by  $n_{j,t}^f = \ln(\ell_{j,t})$ ; using formal firms' first order condition and the law of motion for the log-productivity [\(9\)](#), one can write

$$n_{j,t}^f = \gamma_0 + \gamma_1 \log(\theta_{j,t}) - \gamma_2 w_t + m_{j,t} \quad (31)$$

where  $w_t$  denotes wages,  $\gamma_0 = \frac{1}{1-\alpha} \left[ \log(\alpha) + \log\left(\frac{1-\tau_y}{1+\tau_w}\right) \right]$ ,  $\gamma_1 = \frac{1}{1-\alpha}$ , and  $m_{j,t}$  is measurement error.

One can then use [\(10\)](#) to write [\(31\)](#) in its dynamic representation:

$$n_{j,t}^f = b_0 - \gamma_1 w_t + b_2 w_{t-1} + \rho_f n_{j,t-1}^f + \eta_j + e_{j,t} \quad (32)$$

where  $\eta_j \equiv \gamma_1 (1 - \rho_f) \ln \nu_j$ ,  $b_0 \equiv (1 - \rho_f) \gamma_0$ ,  $b_2 \equiv \gamma_1 \rho_f$ , and the error term is given by  $e_{j,t} = \gamma_1 \epsilon_{j,t} + m_{j,t} - \rho_f m_{j,t-1}$ .

The employment process can be represented as a simple AR(1) process with an MA(1) error, where the MA(1) component arises if one allows for measurement error. The final regression estimated is the following:

$$n_{j,t}^s = \rho_f n_{j,t-1}^f + \Gamma \mathbf{X}_{j,t} + \eta_j + e_{j,t}, \quad (33)$$

where  $\mathbf{X}_{j,t}$  denotes a vector of controls in addition of  $n_{j,t-1}^f$  and  $\eta_j$  denotes firm's fixed effect. The vector  $\mathbf{X}_{j,t}$  includes a set of year dummies and the current and lagged log-average wage rate calculated at the 4-digit industry level. This specification,

which includes current and lagged wages, is standard in the empirical literature (e.g. [Blundell and Bond, 1998](#)).

We start by estimating (33) using both a standard OLS estimator and a within-groups estimator. The former is known to be upward biased while the latter is downward biased, and they can thus be used as upper and lower bounds to any consistent estimator. The third model used is the standard first-differenced GMM estimator, which can be subject to finite sample biases (towards zero) when the lagged levels are weak instruments for the first-differenced equation. The fourth model is the system GMM estimator ([Blundell and Bond, 1998](#)), which uses lagged differences as instruments for equations in levels.

For the first-differenced and system GMM models, we consider two scenarios for the error term. The first is no measurement error, which allows for the use of lagged levels dated  $t - 2$  and earlier as instruments for the differenced equations, and lagged differences dated  $t - 1$  and earlier for the level equations. The second allows for measurement error, which implies that the error term in (33) will have a MA(1) structure. In this case one can only use lagged levels dated  $t - 3$  (and earlier) and lagged differences dated  $t - 2$  and earlier as instruments. All GMM regressions consider the log-wage as a predetermined variable and they are estimated using the two-step estimator with the correction for the variance-covariance suggested by [Windmeijer \(2005\)](#). Table E.5 shows the results.

Table E.5: Productivity process estimation

| Dep. Var.: Log-Employment ( $n_{j,t}$ ) |                    |                    |                    |                    |                     |                     |
|-----------------------------------------|--------------------|--------------------|--------------------|--------------------|---------------------|---------------------|
|                                         | OLS                | FE                 | DIFF1              | DIFF2              | SYS1                | SYS2                |
| $n_{j,t-1}$                             | 0.944**<br>(0.000) | 0.497**<br>(0.002) | 0.594**<br>(0.007) | 0.728**<br>(0.011) | 0.713**<br>(0.009)  | 0.921**<br>(0.005)  |
| $\log(w_t)$                             | 0.0030<br>(0.006)  | 0.0100<br>(0.008)  | -0.0920<br>(0.066) | -0.158*<br>(0.062) | -0.210**<br>(0.072) | -0.339**<br>(0.069) |
| $\log(w_{t-1})$                         | 0.006<br>(0.006)   | 0.005<br>(0.007)   | 0.069<br>(0.095)   | 0.054<br>(0.048)   | 0.110<br>(0.069)    | 0.075<br>(0.053)    |
| Obs.                                    | 741,268            | 741,268            | 741,268            | 741,268            | 741,268             | 741,268             |

Notes: DIFF1 and DIFF2 are the difference GMM models without and with measurement error; SYS1 and SYS2 are the system GMM models without and with measurement error, respectively. Significant at \*\*\*1%, \*\*5% and \*10% levels.

The results shown in Table E.5 follow the pattern expected from the standard results in the literature. The OLS result already points to a very persistent series, which is confirmed by the downward bias apparent in the first-differenced GMM model (DIFF) as compared to the system GMM estimator (SYS). [Blundell and Bond](#)

(1998) find similar results when comparing these two estimators using a small sample of British firms. There is also evidence that measurement error is indeed present, as the estimates under the assumption of no measurement error (DIFF1 and SYS1) seem to be substantially downward biased. The Sargan tests for the additional instruments available in the DIFF1 and SYS1 models (relatively to the DIFF2 and SYS2, respectively) strongly reject the validity of these additional instruments, reinforcing the evidence of measurement error (results not reported). That said, the preferred model is the system GMM under the assumption of measurement error (SYS2), which provides a reasonable value for the persistence parameter.<sup>26</sup>

## E.2 Details of calibration implementation

Let the vector of simulated moments is denoted by  $m_S(\Omega; \zeta)$ , where  $\zeta$  denotes the vector of parameters determined in the first step. The calibrated parameters vector is given by

$$\hat{\Omega} = \arg \min_{\Omega} (\hat{m}_N - m_S(\Omega; \zeta))' (\hat{m}_N - m_S(\Omega; \zeta)) \quad (34)$$

We take the observed wages as given. In order to obtain a wage measure that is more consistent with the model, we estimate a log-wage regression controlling for schooling, gender (male dummy), 4-digit industry dummies, state of residence, a dummy for whether the worker holds a formal contract, dummy for whites, age and age squared, tenure in current job and tenure squared. We restrict the sample to employees only (formal or informal), who are 18 to 69 years old, and who have worked at least 20 hours but at most 84 hours (the 99th percentile) in a given week. We use the estimated coefficients to compute the adjusted wage evaluated at the mean of the vector of observables.

We simulate a cohort of 500,000 potential entrants, which we follow until the age of 50 (within this class of models, Sterk et al. (2021) use a cohort of size 100,000, which they follow for 20 years). For each potential entrant, we draw a pre-entry productivity parameter ( $\nu_j$ ) and a sequence of post entry productivity shocks,  $\epsilon_{j,t}$  for  $j = 1, \dots, 50$ . The stochastic components of the model are drawn only once in the beginning of the procedure and are kept fixed throughout. Each potential entrant has an individual pre-entry productivity parameter,  $\nu_j$ , which is a firm-specific intercept in the AR(1) process that the firm faces after entry. We use a fine grid of 101 equally spaced points for the productivity space. We compute a separate transition probability matrix for each point in the grid and also for the formal and informal sectors (which have different persistence,  $\rho_s$ , and shock variance,  $\sigma_s^2$ ). We do so using the method proposed by Tauchen (1986).

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<sup>26</sup>The Sargan test of overidentifying restrictions rejects instruments' validity in all models, with p-values of zero (not reported). However, Arellano and Bond (1991) show in their simulations exercise that the Sargan test rejects too often in the presence of heteroskedasticity, which is confirmed in their empirical application (they estimate a dynamic employment equation very similar to (33)).

### Smooth Policy Functions

Some of the main decisions firms make in this model are discrete, which implies that some of the policy functions will be step functions. This implies that it is not possible to use derivative-based methods, which are faster and more accurate than derivative-free methods or random search algorithms. We therefore use the smoothing function proposed by [Bruins et al. \(2015\)](#) to correct for the choppiness of the policy functions:

$$h\left(\tilde{V}(\beta), c, \lambda\right) = \frac{\tilde{V}_c(\beta)/\lambda}{1 + \sum_k \tilde{V}_k(\beta)/\lambda}$$

where  $\tilde{V}(\beta)$  is the set of payoffs associated to firms' choices, such as whether to transit to the formal sector or not (if the firm is informal).  $\tilde{V}_c(\beta)$  denotes the net payoff of a given choice  $c$  and  $\lambda$  denotes the smoothing parameter. As  $\lambda \rightarrow 0$ ,  $h(\cdot)$  goes to one if the alternative  $c$  provides the highest payoff and zero otherwise. There is a trade-off between bias and smoothness in the choice of the smoothing parameter: a large value for  $\lambda$  provides a smoother objective function, but can lead to biased estimates; a small value reduces bias but increases choppiness. We follow [Altonji et al. \(2013\)](#) and choose  $\lambda = 0.05$ .

### E.3 Relationship between data moments and parameters

We start by discussing what variation in the data allows us to pin down different parameters. The cost function parameters for both margins of informality are identified by the variation in the share of informal firms by firm size (for the parameter  $\varphi_i$ ) and by the behavior of the share of informal workers within formal firms by firm size (for the parameter  $\varphi_f$ ). In the model, the cost functions of both margins are increasing in firms' size, which implies that the average share of informal firms and the average share of informal workers within formal firms will be decreasing in firm size, as observed in the data. The intensity of this negative relationship in the model is governed by  $\varphi_i$  and  $\varphi_f$  and therefore the behavior of these moments in the data provides the relevant information to estimate these parameters.

The death shocks in both sectors determine the relative disadvantage of being informal regardless of firms' productivity, size or age: the higher the informal death shock,  $\delta_i$  (relative to the formal sector's,  $\delta_f$ ), the greater the relative disadvantage of being informal. They are disciplined by the moments that speak to the relative size of the informal sector. The estimated exogenous exit probability in the formal sector is consistent with the exit probability observed in the data among older formal firms.<sup>27</sup> This is reassuring, as older firms (both in the model and in the data) are more stable and less likely to endogenously exit due to a very negative productivity

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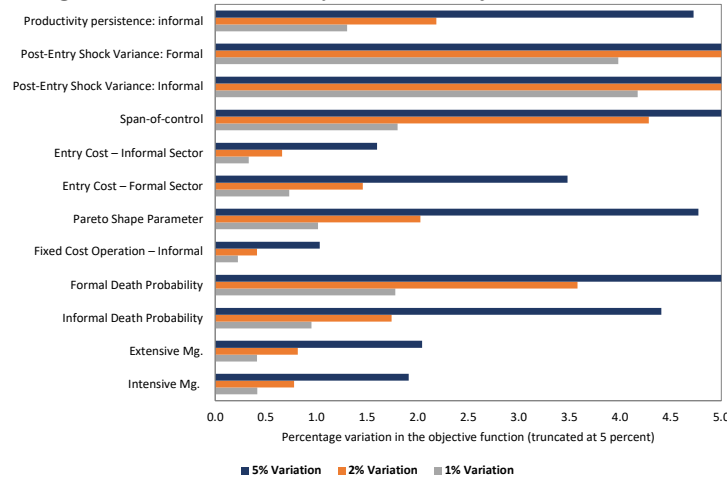
<sup>27</sup>Using a simple linear probability regression conditioning on year and industry dummies, the predicted exit rate at ages 6 to 10 monotonically declines from 0.074 to 0.038.

shock. The entry costs into both sectors are also directly linked to the relative size of the informal sector, but more importantly they directly affect the left tail of the size distribution in both sectors.

The shape parameter of the Pareto distribution,  $\xi$ , is largely determined by the moments of firm size distribution in the formal and informal sectors, and in particular the skewness in formal sector's distribution. The informal persistence and the variance of the shock of the productivity processes in both sectors –  $\rho_i$  and  $\sigma_s^2$ ,  $s = i, f$  – are linked to the moments related to firm growth in both sectors. The  $\sigma_s$  are directly connected to the degree of overlap between formal and informal firm size distributions, as they determine how much productivity dispersion there is conditional to sector choice. These sources of variation in the data contribute to separately estimate the variances and exogenous death shocks.

Finally, we follow [Adda et al. \(2017\)](#) and evaluate if the objective function has curvature around the estimated vector of parameters. If the objective function is flat, then it would suggest that the moments used do not provide relevant information to estimate the parameters. We re-compute the objective function varying each parameter by 1, 2 and 5 percent from its estimated value and compute the corresponding percentage change in the objective function. Figure E.1 shows that there is substantial variation in the objective function in response to small perturbations of the different parameters.

Figure E.1: Sensitivity of the Objective Function



Note: The horizontal bars show the percentage change in the objective function with respect to one, two and five percent changes in the given parameter of the model. The figure is truncated on the right at 5 percent.



## F Computing the transition dynamics

This section presents the details of the computation of the deterministic transition dynamics of the model after the unexpected shock of a permanent increase in labor supply hits the economy. The stationary equilibrium before the shock, in  $t = 0$ , is simply our baseline economy (Brazil in 2003), while the new steady state after the shock is given by the first counterfactual described in Section 5.2 and Table 6. Importantly, the transition dynamics induced by this shock is deterministic (as it is the case with all MIT shocks). The computation algorithm used is the following:

1. Compute the initial and final steady states.
2. Define the formal wage path as  $w_{f,t} = \gamma w_{t-1}$ . Following Schmitt-Grohé and Uribe (2016), our main results use  $\gamma = 0.996$ , which implies that the final steady state is reached in 10 periods. As a robustness check, we also use  $\gamma = 0.997$ , which implies that the final steady state is reached in 15 periods and results remain the same (available upon request).
3. We assume that informal wages adjust freely at all times. Given that we have the value functions at the new steady state in  $t = T$ , we can solve the model backwards from  $t = T - 1, \dots, 1$ . For each period, we solve for the informal wage,  $w_{i,t}$ , that clears the labor market.
4. Simulate the model forward from  $t = 1, \dots, T$  using the sequences of formal and informal wages, and value functions computed in the previous step.
5. Compute the maximum difference between total labor demand and (fixed) labor supply in all periods,  $\Delta = \max |(LD_t - LS)/LS|$ . If  $\Delta$  is small enough, stop.
6. Otherwise, update the guess for informal wages and go to step 3.